

A label-coordinate bookkeeping for entangled pairs: what co-location can and cannot explain

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17 September 2026 · [doi:10.5281/zenodo.23249337](https://doi.org/10.5281/zenodo.23249337) · [CC BY 4.0](#)

Abstract

We formalize and evaluate the conjecture that the non-locality of an entangled pair’s state update belongs to the laboratory’s description: in a chart adapted to the pair, built from a label coordinate and label-dependent shears, the partners are co-located while negativity-entangled. The central result is negative and exact: a rule acting where the partners meet can produce every one-way-no-signaling correlation, so co-location explains neither the Tsirelson bound nor no-signaling; the correlation strength is imported from quantum mechanics, and the one axiom the geometry can be given, order-independence, narrows the class exactly to the no-signaling polytope. The mechanism is stated in four postulates (a locality-graph rule, a state-driven-component rule, a transfer rule for swapping, and an invariant interval clock for a Lorentz-covariant variant in $1 + 1$ dimensions) and proved empirically equivalent to the nonrelativistic quantum mechanics of distinguishable massive particles for spin instruments that do not couple spin to position. Three stronger readings are excluded by the no-communication theorem, by gravity and collider data, and by fine-tuning; a massive-particle test of the first, needing of order 10^2 – 10^3 trials, is proposed. The covariant variant orders a pair’s measurements by a strict Lorentz-invariant partial order carried by events, not charts, undefined on a tie set the symmetric Bell configuration occupies; swapped pairs are at constant separation, not co-located. It differs empirically only from dynamical-collapse models, predicting zero collapse noise as every no-collapse account does. The claim is validity, not correctness: a state-dependent bookkeeping that locates and orders the update, at stated costs.

Keywords: entanglement; Bell nonlocality; interpretations of quantum mechanics; reference frames; relativistic quantum information

1 Introduction

The conclusion first, in plain terms. Two entangled particles that fly apart in the laboratory can be given coordinates in which they never separate, and in those coordinates the update at measurement happens at one point and in a definite order. That is all the construction of this paper delivers, and we prove that it cannot deliver more: a rule acting at the shared point could produce correlations stronger than quantum mechanics allows, so the strength of the quantum correlations, and even the impossibility of signaling, have to be imported rather than explained. Changing coordinates relocates the puzzle of entanglement; it does not remove it. It replaces “instantaneous, at a distance, in no particular frame” by “located, ordered, invariant” — progress on the **distance** in “spooky action at a distance”, not on the **action**: the account says where and when the pair is correlated, not how strongly. The claim is validity — every established result reproduced, no contradiction — not correctness, which no experiment can settle among interpretations that agree on the statistics (Sec. 2).

When one particle of an entangled pair is measured, the joint state of the pair changes at once, however far apart the two particles are. Nothing can be signaled by this [1], but the correlations that result are stronger than any local account with pre-set values allows [2,3], and experiments have confirmed them with every known loophole closed [4–6]. In the laboratory’s description the event is non-local: the state of a distant particle changes because of something done here.

This paper studies a hypothesis about that description — the present author’s own conjecture, formalized and evaluated here for the first time: that the non-locality belongs to the laboratory’s description and not to the particles, because in coordinates adapted to the pair the two are at the same place while entangled and the change at measurement is zero-distance there. To make the hypothesis definite, the particles are given an additional coordinate, the label coordinate z — a label and nothing more, as Sec. 3.6 establishes — with the

laboratory at $z = 0$ and the two members of an entangled pair at $z = \mp 1$. Frames are related by a shear in the laboratory time t whose strength depends on z , so that the same pair of worldlines that the laboratory sees separating at $\pm v$ is seen from either particle’s frame as a single worldline at rest. We call the account that results the **label-coordinate interpretation**, and refer to it below as the mechanism. Two qualifications travel with “at the same place while entangled”: partners paired by entanglement swapping are at constant separation rather than co-located, and in the covariant variant of Sec. 3.7, whose chart updates are retarded, co-location holds up to a light-crossing lag in both directions.

Two honest qualifications govern every locality claim in this paper. First, the “frames” in question are fixed by the entanglement structure of the quantum state: a coordinate choice that depends on the state is not a reference frame in the kinematic sense, and co-location in it is a property of a **state-dependent label chart**, not a demonstration of locality — no dynamics is formulated in the particle charts, and the update rule remains projection, which conditions one wing’s state on the other wing’s spacelike-separated setting, the very dependence Bell’s theorem shows no local account can remove [2]. What the construction does supply is weaker but exact: a **place** (the one chart point both particles occupy at the onset of the first measurement of a pair born together), an **invariant order** of the two measurement events (in the covariant variant, a strict partial order anchored to the pair’s defining event, carried by the events themselves, written into no chart, and undefined on a tie set the symmetric Bell configuration occupies), and a bookkeeping in which the update is zero-distance. Second, in the Galilean form the frame maps never touch the spin state: a single particle’s spatial superposition remains extended in every frame (the maps displace phase-space means only, Sec. 3.5), so the slogan that “the wavefunction does not extend across space” is earned only for the relative coordinate of an entangled pair. What would earn stronger language — a local generator of the update in the particle charts — is examined in Sec. 4.1 and deflates: the standard operational reading of projection already implements the update by an instrument on the measured wing’s tensor factor alone, in every chart, and the label chart adds only that this factor and the partner coincide at one chart point.

1.1 The mechanism in outline

The mechanism has six parts, each developed in Sec. 3; the canonical statement of the form that survives the evaluations is the box at the end of Sec. 3.3, to which every later section refers. **Frames.** Every frame k — the laboratory’s, or a particle’s — carries a frame potential¹ $\sigma_k(z)$; the map from frame k to frame l at label height z is

$$x' = x + [\sigma_l(z) - \sigma_k(z)] t, \quad t' = t. \quad (1)$$

Writing the shear as a difference of potentials makes the composition of frame changes automatic, and $\sigma_k(z_j)$ is the velocity at which frame k sees a particle sitting at height z_j . **Locality graph.** For any number of particles the potentials are fixed by one rule (P1): a frame sees every other particle at its ordinary relative velocity, except its entangled partners, which it sees at rest; partial entanglement is partial locality, the partner seen at a fraction $1 - w$ of its relative velocity, with the weight w of an edge the minimum negativity over the cuts separating its ends, normalized to $[0, 1]$ (P2). **Graph dynamics.** At every time the graph is the entangled-component structure of the state: edges appear at entangling events and disappear with the components themselves, and a destructive Bell-state measurement on two partners transfers locality to the particles they were paired with (P3); in the Galilean form positions integrate the shears and are continuous. **State and collapse.** The spin state is standard quantum mechanics; collapse is projection, and only the **first** projective measurement on an entangled particle changes its partner’s conditional state. Nothing in the frame maps acts on the spin state, the settings or t , which is why the statistics are those of quantum mechanics (Proposition 1). **Positions.** The laboratory’s tracks are the conditional means of continuously monitored wave packets; the frames are attached to those tracks, and for position-entangled pairs the common frame is that of the relative coordinate. **Covariant variant.** Replacing the shear by a z -dependent Lorentz boost — developed in $1 + 1$ dimensions, for massive particles — gives partners a shared rest frame and orders a component’s measurement events by an invariant clock anchored to the component’s defining event (P4).

1.2 What the paper establishes

Three things. First, the mechanism is empirically equivalent to the nonrelativistic quantum mechanics of finitely many distinguishable massive particles, for arbitrary sequences of spin measurements through instruments, local

¹The term is ours; it is unrelated to the “frame potential” of quantum information theory.

to one particle or to co-located particles, that do not couple spin to position (Proposition 1, Sec. 3.1); we restrict the validity claim to exactly that scope wherever it appears. Within it the mechanism charts the co-located matter-system Bell tests — trapped ions [7] and silicon quantum dots [8] — between preparation and readout (the entangling gates and the spin-to-charge readout couple spin to motion and lie outside the instrument class, so no experiment is charted end to end); the rest of the record we cite, photonic or entangled through a photon-mediated step [4,9,10], is consistent by inheritance only: quantum mechanics untouched, no chart supplied. Second, three stronger readings of the hypothesis are confronted with data (Sec. 3.3): an influence that persists after projection is excluded by the no-communication structure of quantum mechanics, and a direct massive-particle test needing of order 10^2 – 10^3 trials is proposed; a physical extra dimension in which electrons move is excluded for universal readings by sub-millimeter gravity and collider data; and a classical rule acting where the particles meet is excluded for its generic members, surviving only in a fine-tuned member that copies quantum mechanics. That last analysis carries the paper’s central negative result: co-location admits every correlation of the one-way-no-signaling polytope, up to the Popescu–Rohrlich value $|S| = 4$ [11], so the geometry explains neither the Tsirelson bound $2\sqrt{2}$ [12] nor even no-signaling. Third, the mechanism differs empirically from exactly one family of rivals, the dynamical-collapse models [13–15], through a prediction of exactly zero collapse noise that it shares with every no-collapse account (Sec. 3.9).

Notation. z is the label coordinate; spin axes $\mathbf{n}$ or $\hat{x}, \hat{y}, \hat{z}$ are unrelated to it. $\sigma_k(z)$ is the frame potential of frame k (Pauli operators appear only as σ or $\sigma_{x,y,z}$), $\beta_{kl} = \sigma_l - \sigma_k$ the shear, w_{ij} the locality weights, w_0 a wave-packet width, $\mathcal{N}$ the negativity, $\mathcal{C}$ the concurrence, N a particle number or trial count, S the CHSH value, λ and r_C the collapse rate and length, λ_Y a Yukawa range, v the electrons’ speed, t laboratory time, τ proper time, κ the ordering clock of P4, and Γ_m the measurement strength of Sec. 3.5.

2 State of the art

2.1 Interpretations and the horn each takes

Einstein, Podolsky and Rosen argued from the correlations of a separated pair that the wave-function description is incomplete [16]; Schrödinger named the phenomenon entanglement [17]; Bohr replied that complementarity undercuts the EPR reality criterion [18]. Bell showed that no theory in which each outcome is fixed by variables local to the measured particle reproduces the singlet statistics [2], Clauser, Horne, Shimony and Holt cast this as $|S| \leq 2$ [3], and quantum mechanics reaches the Tsirelson bound $|S| = 2\sqrt{2}$ and no more [12] — not a consequence of no-signaling alone, since no-signaling correlations reach $|S| = 4$ [11]. Conversely the correlations carry no message [1], which is what lets incompatible accounts of the same statistics coexist.

What separates the interpretations is their answer to the measurement problem, in Maudlin’s decomposition [19]: the linear unitary dynamics, the completeness of the wave-function description and the occurrence of a single definite outcome cannot all hold, so every account gives up one of the three, and Bell’s demand was that the resulting theory be stated exactly, without “measurement” as a primitive [20]. The orthodox (Copenhagen) account keeps a single outcome and treats projection as a change of description, never saying where the cut goes [18,20]. Everett keeps unitarity and completeness and gives up the single outcome [21,22] (decoherence explains einselection but selects no outcome [23,24]; distinguishing Everett from collapse is possible in principle, not with current technology [25]). De Broglie–Bohm keeps unitarity and single outcomes and gives up completeness: definite positions guided non-locally [26–28], with relativistic versions needing a foliation, possibly determined by the wave function [29]. Dynamical collapse keeps completeness and single outcomes and modifies the dynamics — GRW [13], CSL [14,30], gravity-based reduction [31,32] — predicting heating, diffusion and radiation [15]: rival theories rather than interpretations. Relational quantum mechanics [33], QBism [34], consistent histories [35,36] and the transactional account [37] keep the statistics and relocate the meaning of the state. ER=EPR [38], with Susskind’s own generalization to entanglement at large [39], is the only account surveyed that gives entangled partners a geometric connection invisible in the laboratory — the closest relative of the present mechanism. Table 1 collects the assignments; the mechanism’s own row is argued in Sec. 4.1.

2.2 Why no experiment of the Bell kind decides among them

Shimony called the testing of conjunctions of metaphysical theses by Bell-type inequalities experimental metaphysics [40,41], and its record has one pattern: each experiment excludes a class of theories whose predictions differ from quantum mechanics, and none chooses between accounts whose predictions coincide. Excluded so far are local realism; Leggett’s crypto-nonlocal realism [42–44]; macrorealism, up to the clumsiness loophole [45,46];

account	unitary dynamics	complete- ness	single outcome	by what means
orthodox (Copenhagen) [18]	given up	kept	kept	primitive projection; where the cut goes is not said [20]
Everett [21,22]	kept	kept	given up	all outcomes occur, in decohered branches
de Broglie–Bohm [26,27]	kept	given up	kept	definite positions added, guided non-locally
dynamical collapse [13,14]	given up	kept	kept	physical stochastic reduction — a rival theory, testable
relational, QBist, histories, transactional [33–35,37]	kept	reread	reread	what “state” and “outcome” mean is relativized; the statistics are untouched
label-coordinate interpretation (this paper)	given up	kept	kept	primitive projection for the state, as in the first row; decoherence for the charts, read off the conditional state, with a jump at a Bell-basis projection (Sec. 4.1)

Table 1: The horn each account takes in Maudlin’s decomposition [19]. “Reread” marks accounts that keep the statistics and relativize the meaning of the state or the outcome rather than giving up a horn outright.

ψ -epistemic models with independent preparations [47,48], any extension of quantum theory that predicts better than it does, given free settings [49], and non-contextual value assignments [50] (the mechanism adds no outcome-determining variables, so its compliance is automatic); finite-speed influences in a preferred frame, which would permit signaling [51] and are bounded four orders of magnitude above c for most frames [52–54]; and, on the one open front, large parts of the collapse-model parameter planes [55,56] — where collapse models also pose the relativistic question acutely, Tumulka’s flash ontology needing no foliation [57] and Bedingham et al. defining the collapsed matter density on the past light cone [58]. The extended Wigner’s-friend theorems [59–61] exclude conjunctions (absoluteness of observed events, locality, free choice) that interpretations survive by giving up different members; the mechanism keeps absoluteness and free settings and pays on locality, as its Bell accounting does (Sec. 3.2). Where statistics coincide the case has been made on other grounds [62–65], and a poll at a foundations conference found no majority for any interpretation [66]. The mechanism leaves the spin state, the settings, the Born rule and projection untouched (Proposition 1), so the record above applies to it in full, and its one distinct prediction, the absence of collapse noise, is shared with every no-collapse account (Sec. 3.9).

2.3 Nearest neighbors of the co-location idea

The idea that quantum non-locality belongs to the description rather than to the world is not new; what the mechanism may claim as its own is the machinery, not the sentiment.

Quantum reference frames. Giacomo, Castro-Ruiz and Brukner showed that entanglement and superposition are frame-dependent under quantum reference frames: a change to the frame of a quantum system is a unitary, state-independent transformation on an enlarged Hilbert space, under which a state entangled in one frame can be a product in another [67]. This is the mainstream, worked-out version of “non-locality belongs to the frame”, and the contrast is sharp: the QRF transformation is unitary and acts on the state; the label shear is classical, acts on nothing in Hilbert space, and is **state-dependent**, its very existence fixed by the entanglement structure it describes. Four branches of the program bear on this paper: the perspective-neutral framework, in which a frame change is a constraint reduction rather than a fixed unitary [68]; QRF covariance, which makes the partition into subsystems, and with it entanglement, frame-dependent [69]; Lorentz and rest-frame QRFs, where the Wigner rotations that make a wave packet’s reduced spin non-covariant are handled operationally [70–72] — the issues that bound our covariant variant’s scope; and the conservation of entanglement plus coherence under QRF transformations [73], whereas the label chart relativizes the distance between the partners and conserves every quantum measure. The honest comparison is the one Sec. 3.5 makes exact: the label maps displace phase-space means and leave the covariance matrix, so “co-located” in the chart concerns classical conditional means only, whereas in a QRF co-location would mean a relative-coordinate wavefunction peaked at zero, which the label chart never produces (Sec. 4.1).

Deutsch–Hayden; parallel lives; physical influences; retrocausality. Deutsch and Hayden formulated quantum mechanics in the Heisenberg picture so that each system’s descriptors carry all information locally [74]; Wallace and Timpson argued the resulting locality to be a gauge artifact [75], a diagnosis contested since [76–78]. The mechanism makes no Hilbert-space locality claim: its locality object is a classical chart. Brassard

and Raymond-Robichaud constructed a local-realistic account covering every no-signaling correlation [79] — direct prior art for the lesson of Proposition 2: local or geometric constructions of this kind overshoot the quantum set and explain its correlations only by importing them. Pettini’s extra **temporal** dimension [80] is a physical-influence model facing the theorem of Bancal et al. [51], the argument behind this paper’s own exclusions; the mechanism’s coordinate is a label in which nothing propagates. The closest living relative of the paper’s ambition is the retrocausal program surveyed by Wharton and Argaman [81]: Bell correlations carried by parameters on the particles’ worldlines, Lorentz-covariantly, at the price of future-input dependence; both programs refuse a preferred foliation and anchor their story to worldlines and events, and each pays openly — the present mechanism with state-dependent charts and an imported correlation strength. Wave-function realism [82,83], ‘t Hooft’s cellular automata [84] and Kastner’s possibilist transactions [85] rework the same territory.

What the mechanism claims as novel is its machinery and its audit: the locality graph carried by frame potentials with the continuous rule $\ell = 2\mathcal{N}$, the invariant interval clock and retarded chart dynamics (Propositions 4 and 5), the exact characterization of what contact geometry can and cannot supply (Propositions 2 and 3), and the confrontation of the hypothesis’s stronger readings with data (Sec. 3.3). The bare idea that the non-locality is descriptive has at least six prior homes.

3 Analysis

3.1 The frame construction and empirical equivalence

The construction. Every event carries three coordinates: x , the position along the axis on which the pair separates; t , the laboratory time; and z , the label coordinate. Three frames are in play, electron A’s, electron B’s and the laboratory’s, and each frame k is assigned one function of the label, its frame potential $\sigma_k(z)$. The map from frame k to frame l is

$$x' = x + [\sigma_l(z) - \sigma_k(z)]t, \quad t' = t, \quad z' = z, \quad (2)$$

with shear velocity $\beta_{kl}(z) = \sigma_l(z) - \sigma_k(z)$. (When the potentials change with time — the locality weights of Sec. 3.6 evolve under decoherence — the map applies in the integrated form $x' = x + \int_0^t \beta_{kl}(z, t') dt'$, to which every statement below extends.) At fixed z this is a Galilean boost with relative velocity $\sigma_k(z) - \sigma_l(z)$; the novelty is only that the boost velocity may differ from one value of z to another. Composition is automatic, $T_{lm} \circ T_{kl} = T_{km}$ and $T_{lk} = T_{kl}^{-1}$, because differences of potentials telescope. In the gauge $\sigma_i(z_i) = 0$, $\sigma_k(z_i)$ is the velocity at which frame k sees a particle sitting at height z_i .

The particles are placed at $z_A = -1$ and $z_B = +1$ by the event that entangles them, at which both are at $x = 0$, $t = 0$ in the laboratory; the laboratory and everything macroscopic sit at $z = 0$. The laboratory sees the pair separate on $x = \mp vt$, and in either particle’s frame both particles remain at $x = 0$. These nine statements (three objects seen from three frames) fix each potential at $z \in \{-1, 0, 1\}$ and nowhere else; the lowest-degree interpolant is

$$\sigma_A(z) = v(1 - z^2), \quad \sigma_B(z) = -v(1 - z^2), \quad \sigma_{\text{lab}}(z) = vz, \quad (3)$$

and only the values at the three heights ever enter a prediction (Sec. 3.6). In the laboratory the electrons recede from each other at $2v$; in A’s frame B is at rest at A’s position, because $\sigma_A(+1) = 0$, and symmetrically in B’s frame; the laboratory is seen moving at $+v$ by A and at $-v$ by B. At time t the laboratory measures a separation $2vt$ while in either particle’s frame it is 0; Eq. (2) sends one to the other because it boosts the two heights by different velocities. The map between the particles’ frames is itself a shear, $\beta_{AB}(z) = -2v(1 - z^2)$, vanishing at $z = \pm 1$: A and B agree on each other’s position and disagree only about the laboratory. Fig. 1 shows the worldlines.

The state and the measurement rule. The spin state is a density operator ρ on $\mathbb{C}^2 \otimes \mathbb{C}^2$, the singlet $(|01\rangle - |10\rangle)/\sqrt{2}$ at creation. A measurement of $\mathbf{n} \cdot \boldsymbol{\sigma}$ on one particle at laboratory time t_m applies the projection $\rho \mapsto P\rho P / \text{tr}(P\rho)$, $P = \frac{1}{2}(1 \pm \mathbf{n} \cdot \boldsymbol{\sigma})$, with Born-rule probabilities. The frame maps act on x only, never on ρ , on the measurement axes or on t . They are applied to the classical tracks, the conditional-mean positions of Sec. 3.5; the motional state at each height is transformed by the standard Galilean unitary for that height’s boost velocity. The potentials are fixed by the entanglement graph, that is, by the state (postulates P1 and P2 of the canonical box, end of Sec. 3.3); the proposition below uses only one property of those rules, that

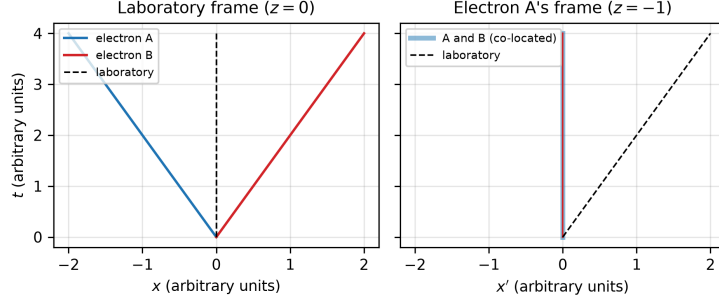


Figure 1: Worldlines of the two electrons and the laboratory ($v = 0.5$). Left: laboratory frame, the pair separates on $x = \mp vt$. Right: electron A's frame, A and B share the worldline $x' = 0$ and the laboratory recedes on $x' = vt$.

they change the potentials and nothing else. Because t is untouched, which of two measurements is first is the same in every frame, and the first projective measurement on the pair is the one that changes the partner's conditional state, at the one chart point both occupy; after it the state is a product state (the variant in which later measurements go on changing the partner is excluded in Sec. 3.3).

Proposition 1 (empirical equivalence). Consider N distinguishable massive particles with spin state ρ and a motional state that factorizes from it, each particle carrying a height $z_i \neq 0$, frames related by Eq. (2), and any finite sequence of quantum instruments on the spin degrees of freedom — completely positive maps with classical outcomes, projective or not, each supported on one particle or on a set of co-located particles — applied at chosen laboratory times. Then the joint outcome distribution predicted by the mechanism equals the standard quantum-mechanical prediction, in every frame.

Proof. (i) The maps Eq. (2) act on x alone, so the sequence of instruments applied to ρ , and their order, fixed by t , are frame-independent. (ii) The outcome rule is the standard instrument rule, so the joint distribution is the standard one; nothing uses projectivity. (iii) The rules of the canonical box change only the potentials, and by (i) a change of potentials cannot change a spin statistic. (iv) Instruments on different particles act on different tensor factors and commute, so the joint distribution does not depend on their time order; the absolute ordering the mechanism supplies has no statistical consequence. (In the covariant variant the same step applies: whenever P4's order differs from a laboratory's coordinate order the two measurements are on different particles — a same-particle sequence is timelike and keeps its order in every frame — so their instruments commute.) (v) Position monitoring acts on the motional state by a completely positive map that is the identity on spin, and spin instruments leave the motional state unchanged. $\square$

The hypotheses delimit the claim, and we hold the paper's language to them: the proposition covers the nonrelativistic quantum mechanics of finitely many **distinguishable massive** particles with spin measured through instruments, each local to one particle or to co-located particles, that do not couple spin to position. Photons are outside it (the covariant variant can order but not locate them, Sec. 3.7), identical-particle geometries are outside it (“which electron carries $z = -1$ ” is not well posed there), joint instruments on separated particles are outside it (Sec. 3.7.3), and real spin readout that proceeds via spin–position coupling sits at its edge (Sec. 3.5). Three corollaries are used throughout: no-signaling holds, since it holds for quantum mechanics; the CHSH value is bounded by $2\sqrt{2}$, but the mechanism inherits rather than derives the bound (Sec. 3.2); and the mechanism is empirically equivalent to every interpretation that reproduces the standard statistics, differing from dynamical-collapse models exactly where those modify them (Sec. 3.9).

Numerical checks. The construction was verified at the standard CHSH settings: $|S| = 2\sqrt{2}$, no-signaling marginals at $1/2$, and frame invariance of S , of the box built by projecting in the absolute time order, and of which measurement is first — across the three frames, 35 combinations of measurement times and laboratory velocities and 200 random measurement-time pairs, with differences in S and in the box probabilities at the 10^{-12} level, composition identities to 10^{-14} (Table 5).

Comparison with experiment. Because the statistics are the standard ones, the mechanism reproduces every Bell-test result that quantum mechanics reproduces; Table 2 lists the benchmarks against which this was checked, with the scope of each. The tightest is the approach to the Tsirelson bound with photon pairs, $S = 2.82759 \pm 0.00051$, 1.6 standard deviations below $2\sqrt{2}$ and never above it [86]; the loophole-free tests are those with nitrogen-vacancy spins 1.3 km apart [4], with photons [5,6], with single atoms 398 m apart [9], and with

superconducting circuits over a 30 m cryogenic link [10]; the co-located matter systems are two beryllium ions in one trap [7] and two electron spins in silicon quantum dots about 60 nm apart [8]. In the nitrogen-vacancy, rubidium and circuit experiments the spins were entangled through photons, a step outside the mechanism’s scope (Sec. 3.6); those data are used only for the statistics they measure.

experiment	system	reported	mechanism	scope
[86]	photon pairs, tabletop	$S = 2.82759 \pm 0.00051$	$S \leq 2\sqrt{2}$	consistency only (photons)
[4]	NV electron spins, 1.3 km, heralded	$S = 2.42 \pm 0.20$	$2 < S \leq 2\sqrt{2}$	consistency only (photon-heralded)
[5]	photons, loophole-free	$p \leq 3.74 \times 10^{-31}$ (11.5 s.d.)	violation	consistency only (photons)
[6]	photons, loophole-free	$p = 5.9 \times 10^{-9}$ (2.3×10^{-7} adjusted)	violation	consistency only (photons)
[7]	two $^9\text{Be}^+$ ions, 3 μm	$S = 2.25 \pm 0.03$	$2 < S \leq 2\sqrt{2}$	in scope between preparation and readout (ions do not separate: trivial chart)
[8]	two electron spins, Si quantum dots, 60 nm	$S = 2.731(88)$	$2 < S \leq 2\sqrt{2}$	in scope between preparation and readout
[9]	two ^{87}Rb atoms, 398 m, heralded, loophole-free	$S = 2.221 \pm 0.033$; no-signaling P -values 0.48, 0.39, 0.014, 0.89, ...	violation; marginals $\frac{1}{2}$, no shift	consistency only (photon-heralded)
[10]	superconducting circuits, 30 m, loophole-free	$S = 2.0747 \pm 0.0033$ (> 22 s.d.)	$2 < S \leq 2\sqrt{2}$	consistency only (photon-mediated link)

Table 2: Benchmarks reproduced by the base mechanism. “Mechanism” is the prediction of Proposition 1, identical to quantum mechanics. Scope: “in scope between preparation and readout” means the mechanism supplies a chart for the interval in which the entangled matter pair exists — the entangling gate couples spin to motion and the readout converts spin to charge or fluorescence, both outside Proposition 1’s instrument class, so no experiment is charted end to end; “consistency only” means the statistics are inherited while the experiment itself lies outside the mechanism’s scope. The P -values of [9] test each wing’s outcome distribution across the remote setting; among eight such tests (four data sets, both directions), a single $P = 0.014$ is not significant after a trials correction.

The result is consistency, and nothing more: no spin statistic separates the base mechanism from quantum mechanics. What it adds is a place where the pair’s state changes, and a reason the laboratory reports the change as instantaneous — its chart is sheared relative to the pair’s. The geometry says where and when the correlation is established; the next subsection asks whether it also fixes how strong it is.

3.2 Can co-location explain the correlations? The contact-rule test

What was evaluated. The strongest reading of the hypothesis. “Measuring one particle immediately changes the other because they are local” can be read as a claim about a real, classical interaction at the point the two particles share in their own frames: the first measurement (first in laboratory time, which every frame agrees on) fixes A’s outcome, and B’s outcome is then fixed by a rule that may use everything present at the contact point — A’s setting, A’s outcome and B’s own setting. If such a rule produced the singlet correlations, respected no-signaling and stopped at the Tsirelson bound without being copied from quantum mechanics, the geometry would account for the strength of the correlations and not merely host them.

How. Each wing has two settings, $s, s' \in \{0, 1\}$, and two outcomes, $a, b \in \{+1, -1\}$. A contact model is any joint distribution of the form

$$P(a, b \mid s, s') = P_A(a \mid s) P_B(b \mid s, a, s'), \quad (4)$$

in which B’s rule may depend on s and a because, at the contact point, both are available; a Bell-local model would allow only $P_B(b \mid s', \xi)$ with a shared variable ξ fixed at the source. In the classification of Jarrett and Shimony [87,88], Eq. (4) is the parameter-independence / outcome-independence decomposition of Bell’s factorizability: the wing measured second may depend on the first wing’s **outcome** and, in the unrestricted class, on its **setting**; Wiseman’s account of which locality notion each of Bell’s two theorems targets is the modern statement [89]. Every model of the form Eq. (4) was evaluated on the CHSH value $S = E_{00} - E_{01} + E_{10} + E_{11}$ and on a signaling measure, the largest change of one wing’s marginal when the other wing’s setting is changed. We enumerated every deterministic strategy — $a = f(s)$ has 4 choices and $b = g(s, a, s')$ has 256, hence 1024 — then stochastic rules, then the quantum-tuned rule with P_A the singlet’s marginal and P_B its conditional.

Proposition 2 (contact rules are exactly the one-way-no-signaling boxes). A joint distribution $P(a, b \mid s, s')$ admits the form Eq. (4) if and only if A’s marginal is independent of B’s setting s' . *Proof.* Any distribution factorizes by the chain rule on the support of P_A , $P(a, b \mid s, s') = P_A(a \mid s, s') P_B(b \mid a, s, s')$; if the A-marginal is s' -independent this is Eq. (4), and conversely. $\square$

Contact structure therefore characterizes exactly the polytope of **one-way-no-signaling** boxes — those that cannot signal from the second wing to the first. That set contains the entire no-signaling polytope, the Popescu–Rohrlich box [11] included, and beyond it every box that signals from the first wing to the second (Bancal et al. build their finite-speed-influence models from precisely such one-way boxes [51]). Both structural features of the quantum correlations — the Tsirelson bound **and** no-signaling toward the unmeasured wing — are imported from quantum mechanics, not supplied by the geometry. We state this as a general caution: any account that explains entanglement by giving the partners a geometric connection with an ordered meeting point — the present mechanism, and informal readings of ER=EPR [38] among them — inherits the correlation strength from quantum mechanics, because the geometry admits strictly more than quantum mechanics allows. (That a local-realistic construction can fill the whole no-signaling set was shown by Brassard and Raymond-Robichaud [79]; Proposition 2 locates the boundary for contact geometries one step further out.)

Proposition 3 (order-independence narrows the polytope). In the mechanism the measurement order is fixed by times and worldlines and varies from run to run. If the box is required not to depend on that order — an input from quantum mechanics and from experiment, not a consequence of the geometry, which would host an order-dependent box as readily — then the same box must be realisable as a contact rule with **either** particle measured first. A box is realisable with A first iff its A-marginal ignores s' (Proposition 2), with B first iff its B-marginal ignores s ; both together are exactly no-signaling. *Order-independence therefore cuts the one-way-no-signaling polytope exactly down to the no-signaling polytope.* It cuts no further: the PR box is order-independently realisable, so the Tsirelson bound remains out of reach. What **does** select the quantum boundary has its own literature — information causality [90], macroscopic locality [91], local orthogonality [92] — and Proposition 3 locates the geometry’s entire contribution to that program: one imported axiom, which reaches no-signaling and stops.

Results of the enumeration. A deterministic strategy has every $E_{ss'} = \pm 1$, so $S \in \{0, \pm 2, \pm 4\}$. Of the 1024 strategies, 128 reach $|S| = 4$, and every one of them signals; the no-signaling deterministic strategies number exactly 256 and their largest $|S|$ is 2. The reason is short: $b = g(s, f(s), s')$ is a function of (s, s') , so B’s marginal is a point mass that is independent of s only if $b = h(s')$, an ordinary local model bounded by $|S| \leq 2$ [2,3] (16 such strategies, each realized by 16 contact rules differing only in entries never reached): a deterministic contact rule that uses A’s setting always leaks it, and one that does not leak it cannot use it. Stochastic rules are different: let A answer uniformly at random and let B copy a except when $(s, s') = (0, 1)$, where B answers $-a$ — uniform marginals, signaling 0, $S = 4$, the PR box as a contact rule. Since the 16 local strategies and the 8 PR-type boxes are the vertices of the no-signaling polytope [93], every no-signaling correlation can be produced by a rule of the form Eq. (4); the quantum-tuned rule reproduces the singlet box exactly, $S = 2\sqrt{2}$, signaling 0, all sixteen probabilities the quantum ones. In sum: 256 deterministic no-signaling (local) strategies with $|S| \leq 2$, 768 signaling ones of which 128 reach $|S| = 4$, a continuum of stochastic rules whose no-signaling members fill the no-signaling polytope, and one quantum-tuned rule at $2\sqrt{2}$.

Conclusion, and what the data exclude. Co-location supplies two things the Bell argument in spacetime forbids — an absolute order, so that “second measurement” is well defined in every frame, and a place where a conditional rule can act locally on what the first measurement produced — but not the numbers: the value $2\sqrt{2}$ that nature stops at [12] is selected by nothing in the construction and must be imported, as must the conditional probabilities themselves. This is the sense in which Proposition 1’s equivalence is inherited rather than derived. Generic members of the class are excluded; the class as a whole cannot be, because it contains the quantum point. Every deterministic rule with $|S| > 2$ signals, and the explicit no-signaling test of [9] found no evidence of signaling. A generic stochastic rule has no reason to respect the Tsirelson bound: the loophole-free record caps $|S|$ at or below it — 2.0747 ± 0.0033 [10], 2.42 ± 0.20 [4], 2.221 ± 0.033 [9] — and the tightest approach, $S = 2.82759 \pm 0.00051$, three parts in 10^4 below $2\sqrt{2}$ with no run above it [86], though as a fair-sampling tabletop measurement it cannot by itself constrain signaling or super-quantum rules. What remains is disfavored by fine-tuning: to survive, a contact rule must sit exactly at the quantum conditionals. The analysis takes the settings to be free — supported by the human-choice and cosmic Bell tests [94–96] — and the surviving rule is not a superdeterministic escape, using the settings only at the contact point, after they

are chosen. A contested reanalysis of the nitrogen-vacancy data [97] has claimed a no-signaling anomaly; it has not been confirmed, and the dedicated test of [9] found none. The quantum-tuned rule survives as what it is: the standard outcome-dependent, parameter-independent factorization of the quantum box in Jarrett’s terms [87,88], narrated at the contact point — it contains no hidden variable and adds nothing empirical to the base mechanism.

3.3 Variants excluded by existing data

Three readings of the hypothesis say more than the base mechanism of Sec. 3.1. Each was made precise enough to predict something the base mechanism does not, and each prediction was confronted with published data; in each case the reading that adds physics is excluded and only a reading identical to the base mechanism survives. Together they fix the wording of the mechanism that survives.

(a) The persistent-influence variant. In the base mechanism the first projective measurement on the pair changes the partner and nothing after it. A stronger reading keeps the labels in place after projection and lets “measuring one changes the other” go on applying: every later projection of one particle re-projects the partner into the opposite eigenstate of the same operator. This is the only version of the label’s behavior at projection that contains new physics; the inert alternatives — the labels persist with the edge weight at zero, or the projected particle leaves the labeled set — share every prediction. Let A be measured along $\mathbf{n}_1$ at t_1 with outcome s_1 , and let a second round at $t_2 > t_1$ measure A along $\mathbf{n}_2$ and then B along $\mathbf{n}_3$. After the first projection the singlet is the product state in which A is the s_1 eigenstate of $\mathbf{n}_1 \cdot \boldsymbol{\sigma}$ and B the $-s_1$ eigenstate, so standard quantum mechanics gives

$$E_2(\mathbf{n}_2, \mathbf{n}_3) = (\mathbf{n}_1 \cdot \mathbf{n}_2)(-\mathbf{n}_1 \cdot \mathbf{n}_3), \quad (5)$$

whereas the persistent-influence rule sends B to the $-s$ eigenstate of $\mathbf{n}_2 \cdot \boldsymbol{\sigma}$ whenever A obtains s , so that

$$E_2^{\text{infl}}(\mathbf{n}_2, \mathbf{n}_3) = -\mathbf{n}_2 \cdot \mathbf{n}_3, \quad (6)$$

independent of $\mathbf{n}_1$; the difference is $-\mathbf{n}_{2\perp} \cdot \mathbf{n}_{3\perp}$, of magnitude 1 at $\mathbf{n}_2 = \mathbf{n}_3 \perp \mathbf{n}_1$. **Results.** Condition on A’s first-round outcome, say $s_1 = +1$ — a classical record, communicable between rounds. Under the variant, B’s second-round marginal along $\hat{z}$ is $P(b = -1) = 1$ if A re-measured $\hat{z}$ but $\frac{1}{2}$ if A measured $\hat{x}$: a signaling measure of 0.5 in the s_1 -conditioned probability (the unconditional single-shot marginal is uniform for either setting; the signal requires the shared past datum s_1 , classically available before round two, so the protocol signals). The same box at the CHSH settings gives $|S| = 2\sqrt{2}$ on a product state, where quantum mechanics gives at most 2 ($\sqrt{2}$ at the standard settings). The variant therefore contradicts the no-communication theorem [1] — a structural, and structurally verified, feature of quantum mechanics. The atoms of the no-signaling test [9] were each measured once, so its marginal checks constrain the first round only; but the photonic “nonlocality-sharing” experiments, following the theory of [98], measure one wing **twice** — a first measurement of tunable strength ε , then a projective one — against a projective measurement on the other wing, the geometry of Eq. (5) and Eq. (6) for polarization: Schiavon et al. found the second-observer CHSH following the quantum product-state law $\sqrt{2}(1 + \cos \varepsilon)$ through the projective endpoint, where the persistent-influence rule predicts $2\sqrt{2}$ [99], and Hu et al. confirm the law to near-projective strengths [100]. The variant is a rule about labels, and photons carry none, so what the photonic data exclude — by a margin of order unity against per-mille statistics — is the prediction the variant **would** make transposed to polarization; the exclusion that binds the variant itself is the structural one. The inert alternatives are indistinguishable: the laboratory tracks stay $x = \mp vt$ under both (the four potential values that fix them are pinned by Eq. (3)), and in either particle’s frame the partner recedes at $2v$ from the projection event on — with $v = 0.5$ and projection at $t = 0.5$, a separation of 0.5 at $t = 1$ against 1 in the laboratory — whether its label persists with weight zero or is deleted.

A direct in-scope test is a two-basis sequential QND measurement on both particles after the first projection, on a platform that prepares a singlet and reads out both spins in place — two ions in one trap [7,101] (the double-dot readout of [8] is a destructive spin-to-charge conversion, not a per-spin QND readout): measure A along $\hat{z}$, rotate both readout bases to $\hat{x}$, measure A, then B. The observable $E_2(\hat{x}, \hat{x})$ is 0 under quantum mechanics and every inert variant, -1 under persistent influence, and the null is robust to preparation ($\langle \hat{x} \cdot \boldsymbol{\sigma} \rangle_A = 0$ after a $\hat{z}$ projection whatever B’s state). With readout fidelity F per measurement the observed correlation is $(2F - 1)^2 E_2$, and separating 0 from $-(2F - 1)^2$ at 5 standard deviations needs $N(F) = \lceil 25/(2F - 1)^4 \rceil$ trials — 26 at $F = 0.9994$, the repeated QND readout fidelity demonstrated on $^{27}\text{Al}^+ / ^9\text{Be}^+$ [102], 39 at $F = 0.95$, 400 at $F = 0.75$ — where $N(F)$ uses the null-hypothesis variance $1/N$ (conservative) and a large- N approximation

that is only indicative at $N = 26$; the $F = 0.9994$ figure is demonstrated on a spectroscopy ion read through a logic ion, a different platform than the singlet sources, so the combination is proposed, not demonstrated, and at these trial counts systematics (readout crosstalk, the calibration of the $\pi/2$ rotation, the QND-ness of the rotated-basis readout) dominate. Routine sequential two-basis readout on co-trapped ions would show $E_2(\hat{x}, \hat{x}) \approx 0$ as a matter of course, so the run is a formality rather than a discovery. **Conclusion.** The variant is excluded by the no-communication structure of quantum mechanics and, for photons, by the nonlocality-sharing data [99,100]; a direct massive-particle test needing of order 10^2 – 10^3 trials would exclude it on the platforms the mechanism describes. Only the first projective measurement on an entangled pair changes the partner.

(b) The physical-dimension variant. The reading in which the label coordinate is a physical compact dimension of size R that electrons occupy while entangled, displaced from the laboratory by a physical distance L . Write $Z = Lz$; the frame maps keep their form and every spacetime observable is independent of L . On a circle of circumference $2\pi R$ two points at $Z = \mp L$ are at most antipodal, so $2L \leq \pi R$. In the free-motion sub-variant the electron is a field on the circle with quantized compact momentum $p_Z = n\hbar/R$, a Kaluza–Klein tower $E_n \approx n^2\hbar^2/(2m_e R^2)$ for $R \gg \hbar/(2m_e c)$, and a packet of width $\delta \leq R$ placed at $Z = \pm L$ spreads round the circle in $t_{\text{spr}} \lesssim m_e R \delta / \hbar \leq m_e R^2 / \hbar$; in the placement sub-variant the entangling event puts the electron at $Z = \pm L$ with no kinetic term in Z . If the dimension is universal, so that gravity propagates in it, the Newtonian potential at $r \lesssim R$ acquires a Yukawa tail $U(r) = -(Gm_1 m_2 / r)(1 + \alpha e^{-r/\lambda_Y})$ with $\lambda_Y \sim R$. **Results.** The Eöt-Wash torsion balance finds the inverse-square law Newtonian from $52 \mu\text{m}$ to 3 mm , requires any Yukawa term with $|\alpha| = 1$ to have $\lambda_Y < 38.6 \mu\text{m}$ (95 % CL), and bounds the largest extra dimension at $R^* < 30 \mu\text{m}$ [103]; the HUST pendulum finds the inverse-square law holding down to $48 \mu\text{m}$, so a Yukawa term with $|\alpha| \geq 1$ must have $\lambda_Y < 48 \mu\text{m}$ [104]. A gravitationally universal label dimension therefore has $R < 30 \mu\text{m}$ and $L \leq \pi R/2 < 47.1 \mu\text{m}$, inside which the fractional force deviation at 1 mm ($\alpha = 1$, $\lambda_Y = 30 \mu\text{m}$, point masses) is 1.1×10^{-13} , below any tabletop sensitivity. If all Standard-Model fields propagate in it, the collider bounds on universal extra dimensions are overwhelming: $1/R \gtrsim 300 \text{ GeV}$ already from precision electroweak data [105], and $1/R > 1.4$ – 1.5 TeV from current LHC data [106], i.e. $R \lesssim 1.4 \times 10^{-19} \text{ m}$, far below the electron Compton wavelength, where the nonrelativistic spreading formula no longer applies and the only time scale left is the light-crossing time $R/c \approx 5 \times 10^{-28} \text{ s}$. The free-motion sub-variant fails independently of gravity in any case: at the largest allowed radius, $R = 30 \mu\text{m}$, the spreading time is $t_{\text{spr}} = 7.77 \mu\text{s}$, while entanglement between two co-located calcium ions has been certified for 20 s , with a coherence decay constant of $34(3) \text{ s}$ [107], entangled logical states of trapped ions have been stored for about one hour ($3.6 \times 10^3 \text{ s}$) [108], and a decoherence-free-subspace logical qubit — itself a two-ion entangled state — has kept coherence for more than two hours [109]: the ratio of the shortest certified time to t_{spr} is 2.6×10^6 . A freely moving electron would leave $Z = \pm L$, and its frame, long before the pair is measured. **Conclusion.** Two registers must be kept apart. The spreading-versus-storage argument is an **internal-consistency** exclusion, needing no gravity data; the gravity and collider bounds are **external** exclusions binding only the universal readings. The non-universal placement reading — a dimension only the labels inhabit, in which nothing propagates — is constrained by no data at all, and we say so plainly: what survives is exactly the reading with nothing left to measure, the label of Sec. 3.1 with a length attached that nothing measures.

(c) The classical contact rule. The contact-rule class of Sec. 3.2 is the third exclusion, in the class register stated there: generic members are excluded and the sole survivor is the fine-tuned member that copies the standard conditionals exactly and adds nothing.

The mechanism that survives. We now state the surviving mechanism once, canonically.

Canonical statement. Each particle i of the labeled set S carries a label height $z_i \neq 0$, assigned by the event that entangles it, at one place in the laboratory; the laboratory and everything macroscopic sits at $z = 0$. The label is a label, not a place: nothing moves in it, and a particle keeps its height while it belongs to S — after a projection its edges carry weight zero and every frame sees it at its full relative velocity (equivalently, it may be dropped from S ; the two readings share every observable, item (a) above). Every frame k carries a potential $\sigma_k(z)$, frames are related by the shear Eq. (2) (in integrated form when the potentials change with time), and the frames are attached to the conditional-mean tracks of continuously monitored wave packets (Sec. 3.5). The spin state, its unitary dynamics and the Born rule are those of standard quantum mechanics; collapse is projection.

When. Two things change at a measurement, on two clocks. The **conditional state** of S changes by projection at the completion of the measurement, the registration of its outcome — in the base form at its laboratory time, which every frame agrees on — and the first projective measurement on an entangled particle, in its component's order, is the one that changes its partner's conditional state. The **charts** are read off the **outcome-conditioned** reduced state of S , which until the outcome is registered is the dephasing reduced state: the weights fall continuously through the premeasurement, in which the measured wing's interactions with its environment lower the edge weights of P2 (Proposition 4), and the projection completes them — with no jump at a single-particle projection, where the dephased and the projected weights coincide, and a jump from 0 to 1 of the swapped edge of P3 (and, when the measured pair survives, of its own edge) at a Bell-basis projection — so “nothing jumps” holds for single-particle measurements and the decohering stage only; in the covariant variant the change reaches each chart retarded (P4). P4 orders projections by their onset events — the event where the instrument meets the particle's worldline, a decoherence time before the registration and immaterial at every scale computed here — and orders no chart change. Which systems belong to S — the microscopic particles that carry labels, with readout photons, motional modes and charge sensors outside it — is a stipulation of the interpretation, priced in Sec. 4.1.

P1 (locality-graph rule). Each pair (i, j) carries a weight $w_{ij}^{(k)} \in [0, 1]$ as read in chart k , the locality graph. In the base form every chart reads the same graph and the superscript is dropped; in the covariant variant chart k reads the graph off the state on its own past light cone (P4), so inside a light-crossing gap two charts can disagree on an edge — the measured wing's chart has $w = 0$ where its partner's still has $w = 1$ — and the weights are chart-indexed by construction. The potential of particle i is pinned at every height and at the laboratory,

$$\sigma_i(z_i) = 0, \quad \sigma_i(z_j) = (1 - w_{ij}^{(i)})(v_j - v_i), \quad \sigma_i(0) = -v_i, \quad \sigma_{\text{lab}}(z_j) = v_j, \quad (7)$$

and between the pinned heights it is the lowest-degree polynomial through those values: a frame sees every other particle at its ordinary relative velocity, except its partners, which it sees at rest.

P2 (state-driven components). At every time the graph is read off the quantum state: the weight of the edge (i, j) is the minimum negativity over the bipartite cuts separating i from j , normalized by its maximum for the smaller of the two spin dimensions — $w_{ij} = 2\mathcal{N}_{\min}/(d_{\min} - 1)$, which lies in $[0, 1]$ for any spins and is twice the minimum negativity for spin-1/2 — evaluated on the outcome-conditioned reduced state of S (“When” above). The rule is continuous in the state (Proposition 4), reduces to twice the negativity for a lone electron pair, and makes partial entanglement partial locality. Edges disappear at measurements whose post-measurement states are product states, while a measurement in an entangled eigenbasis leaves its measured pair edge-connected. In the covariant variant the graph in chart k at an event is the component structure of the state on that event's past light cone.

P3 (transfer — the destructive special case of P2). A destructive Bell-state measurement on two partners removes their edges and creates an edge between the particles they were paired with; swapped partners are thereby at constant separation — at rest relative to each other, not co-located — without a common creation event (Sec. 3.6). When the measured pair survives in an entangled eigenstate, P2 governs: the surviving pair is itself a new component, alongside the swapped one.

P4 (invariant ordering and retarded charts; covariant variant only). With the shear replaced by a z -dependent boost (Sec. 3.7), each component carries one ordering clock, read at each of its measurement events: the signed invariant interval κ from the component's **defining event** $\mathcal{E}$ — the event at which the component came to be, its creation or an entangling interaction, or the transfer of P3; a component created by pruning inherits the defining event of the component it was pruned from — to the measurement event, i.e. the proper time an inertial clock would accumulate from $\mathcal{E}$ to that event, positive inside $\mathcal{E}$'s future light cone, zero on it (and on the past null cone) and negative elsewhere outside it (Eq. (9)). A particle belongs to successive components, and its measurement event carries one κ per component. The measurements of a component are ordered by the causal order wherever relativity defines one and, between spacelike-separated events inside $\mathcal{E}$'s future cone, by the smaller κ ; events outside that cone are directly ordered only causally (the transitive closure then places such an event before every κ -successor of its in-cone causal successors),

and equal κ leaves a pair unordered. The clock is invariant but not retarded: it may refer to a defining event outside the measurement's own past cone. The charts are retarded: every graph change caused by an event reaches particle i 's chart where that event's future light cone meets i 's worldline, and a chart's potentials are those set at the most recent such crossing of its own worldline — a clause about potentials only, never about the ordering. On this rule a faster particle of a pair born together measures later at equal laboratory times, which is the one place the rule has content; the order is a strict partial order agreeing with the causal order (Proposition 5) — within a component; the orders of different components need not compose (Sec. 3.7.2) — and no chart changes outside the future light cone of the event that causes the change.

Throughout, the spin statistics are those of quantum mechanics, including the Tsirelson bound, which the geometry admits but does not select (Proposition 2); what the mechanism claims for itself is the location and the invariant timing of the change. The remaining subsections test whether that claim survives realistic positions, many particles and relativistic kinematics.

3.4 Degree of entanglement and degree of locality

The frame construction was stated for the singlet. We evaluated whether it extends to partially entangled pairs in a way that makes partial entanglement a partial locality, and whether that extension changes any statistic. The locality-graph rule of Sec. 3.6, Eq. (7), specialized to a pair with weight $w \in [0, 1]$ and lab tracks $x = \mp vt$, pins $\sigma_A(z_A) = 0$, $\sigma_A(z_B) = 2v(1 - w)$, $\sigma_A(0) = v$, $\sigma_{\text{lab}}(z_{A,B}) = \mp v$, and the mirror image for B : a partner of weight w is seen at a fraction $1 - w$ of its ordinary lab relative velocity. Defining the locality fraction ℓ as one minus the ratio of the separation rate in a particle's own frame to the lab's rate $2v$, Eq. (7) gives $\ell = w$ exactly (verified for $w \in \{0, 0.25, 0.5, 0.9, 1\}$). “Degree of locality equals degree of entanglement” holds with the negativity as the measure: for a two-electron spin state ρ the edge rule of P2 gives $w = 2\mathcal{N}(\rho)$, twice the negativity [110,111] — 0 for separable states, 1 for the singlet, and **equal to the concurrence** $\mathcal{C}$ of [112] on both families quantified below (every pure two-qubit state and every Werner state; in general $2\mathcal{N} \leq \mathcal{C}$, a theorem of Verstraete et al. [113], reproduced on 40 random mixed states, so a concurrence-weighted rule would assign a different ℓ off those families — a choice of the same standing as the pinned-lab convention below). On those families $\ell = 2\mathcal{N} = \mathcal{C}$.

The linearity is a convention, and we state the alternative considered. If instead the pair is placed at heights $z = \mp \mathcal{C}$ with the quadratic potentials of the singlet construction, one finds $\ell = \mathcal{C}^2$ — but also that the lab sees the pair separate at $2v(1 + \mathcal{C} - \mathcal{C}^2)$ rather than $2v$: a partially entangled pair would fly apart faster than a singlet from the same source, a kinematic statement no entanglement measure should make. Pinning the lab tracks at $x = \mp vt$ for every $\mathcal{C}$ removes that defect and removes all dependence on how the particles' potentials are interpolated between the pinned heights; the rule Eq. (7) builds the pinned-lab convention in.

For the Werner state $\rho_{p_W} = p_W |\psi^-\rangle\langle\psi^-| + (1 - p_W)\mathbb{1}/4$ the two numbers are $\mathcal{C} = \max(0, (3p_W - 1)/2)$ and $S_{\text{max}} = 2\sqrt{2}p_W$ (Horodecki formula): CHSH is violated only for $p_W > 1/\sqrt{2}$, i.e. $\mathcal{C} > 0.5607$, whereas $\mathcal{C} > 0$ — and hence $\ell > 0$ — already for $p_W > 1/3$, so a pair can be partly local without violating any Bell inequality. For the pure states $\cos\theta |01\rangle - \sin\theta |10\rangle$ the closed forms are $\mathcal{C} = \sin 2\theta$ and $S_{\text{max}} = 2\sqrt{1 + \sin^2 2\theta}$, so every $\mathcal{C} > 0$ violates CHSH: at $\theta = 0, \pi/16, \pi/8, 3\pi/16, \pi/4$ the pairs $(\mathcal{C}, S_{\text{max}})$ are $(0, 2)$, $(0.3827, 2.1414)$, $(0.7071, 2.4495)$, $(0.9239, 2.7229)$ and $(1, 2.8284)$. The two families show that ℓ is not a function of S : the Werner state with $p_W = 0.8$ and the pure state with $\sin 2\theta = 0.7$ share $\mathcal{C} = 0.7$ but give $S = 2.2627$ and 2.4413 ; the same Werner state and the pure state with $\sin 2\theta = 0.5292$ share $S = 2.2627$ but have $\mathcal{C} = 0.7$ and 0.5292 . The locality fraction is fixed by the state, which tomography measures, not by the Bell parameter. As an illustration only, published S values can be read through the Werner assumption $p_W = S/(2\sqrt{2})$: the silicon quantum-dot pair of [8] then moves from $\mathcal{C} = 0.948(47)$ at 0.1 K ($S = 2.731(88)$) to $\mathcal{C} = 0.614(34)$ at 1.1 K ($S = 2.101(64)$, from that paper's extended data) on the same device — from 95% to 61% co-located in the particles' chart, while the lab statistics are just those of the measured state. The reading is not a measurement of $\mathcal{C}$: for the beryllium ions of [7] the implied singlet fidelity falls short of the published $\approx 88\%$.

The statistics are untouched: for 200 random mixed and 200 random pure two-qubit states with $\mathcal{C}$ from 0 to 0.995, the outcome probabilities of the weighted construction are identical to those of quantum mechanics, no-signaling holds to 10^{-16} , and $S \leq 2\sqrt{2}$ throughout. What the evaluation adds is definitional: “degree of locality” becomes a defined number, $\ell = 2\mathcal{N}(\rho)$, computable from any tomographically known state, with no prediction depending on it except through ρ itself.

3.5 Where the particles are: realistic tracking and position entanglement

The frame maps are attached to the electrons' lab tracks $x = \mp vt$. Quantum mechanics forbids tracking a free electron exactly, so we evaluated what “the track” means when the lab monitors each electron's position as well as the uncertainty principle allows, whether that monitoring disturbs the spin singlet, and what “at the same place” means when the positions themselves are entangled [16]. The interpretation adds one rule to both cases: the frames attach to conditional means, and the frame maps displace phase-space means only, leaving every covariance chart-independent.

Continuous weak measurement of the tracks. Each electron's motional state is a Gaussian wave packet under continuous weak measurement of x with strength Γ_m , in the conventions of [114,115] (Appendix A). The paths of the frame construction are the deviations of the conditional-mean tracks from the ideal ones, and the frame maps are unchanged; nothing in the monitoring acts on the spin. Four conclusions carry forward. (i) The monitored packet reaches a minimum-uncertainty steady state with $\sqrt{V_x} = (\hbar/8m\Gamma_m)^{1/4}$; the Heisenberg product decays monotonically to $\hbar^2/4$ and can never be broken (Eq. (11)), while the monitoring heats each electron at $\hbar^2\Gamma_m/m$. (ii) The one kinematic parameter the frame maps need, the separation speed $2v$, is measurable to 0.92% at micrometer resolution ($\Gamma_m = 10^{19} \text{ m}^{-2}\text{s}^{-1}$, $T = 100 \text{ ns}$), optimally 0.81% at $T^* = 37 \text{ ns}$; stronger monitoring makes the track **worse** — at tens-of-nanometer resolution the momentum is randomized within nanoseconds. (iii) Pushed through the frame maps, the tracked paths return co-location up to tracking noise: in A's frame the tracked B has zero-mean separation with rms $17.8 \text{ }\mu\text{m}$ at $t = 100 \text{ ns}$ (Eq. (13)), against a lab separation of 2.0 mm . (iv) The spin is untouched: the measurement term traces out of the spin sector identically ($S = 2\sqrt{2}$ and $\mathcal{C} = 1$ through a tracking run). The physical condition is the absence of a spin–position coupling: with a sensor reading a spin-dependent position (a magnetic-field gradient, operator $A = \sigma_z^A \otimes x$ in the double commutator of Eq. (10)) the reduced spin dephases at $4\Gamma_m\langle x^2 \rangle$ — the two spin branches see $\pm x$ — and one Euler step $\delta t = 0.05/\Gamma_m$ of that dissipator on a discretized position state (four levels, unit strength) already lowers $|S|$ at the standard settings from $2\sqrt{2} = 2.83$ to $\sqrt{2}(2 - 4\Gamma_m\delta t\langle x^2 \rangle) = \sqrt{2}(2 - 0.2\langle x^2 \rangle)$, which is 2.47 for the uniform mixture of the levels $x \in \{\pm 0.5, \pm 1.5\}$ ($\langle x^2 \rangle = 1.25$): the fragility that bounds the honest scope of Proposition 1. Heisenberg-limited position monitoring has been demonstrated on mechanical oscillators [116,117], and charge sensors read which dot each electron occupies while a (1,1) singlet persists [118,119] — the benign case, with spin read out only when Pauli blockade converts it to charge; the precision result applies to free electrons in flight.

Position–momentum entanglement. The motional state along the separation axis was then taken to be a two-mode squeezed Gaussian with parameter $r \geq 0$ on packets of width w_0 (momentum spread $w_p = \hbar/2w_0$), with $\text{Var}(x_1 - x_2) = 2w_0^2e^{-2r}$, $\text{Var}(p_1 + p_2) = 2w_p^2e^{-2r}$ and the conjugate combinations anti-squeezed by e^{2r} ; the state is pure and, for the spin, still the singlet, $\rho_{\text{spin}} \otimes \rho_{\text{motion}}$ — an assumption, not a result. A frame map displaces the phase-space mean, $x_i \rightarrow x_i + \beta(z_i)t$, $p_i \rightarrow p_i + m\beta(z_i)$, and leaves the covariance matrix unchanged: in A's or B's frame both means vanish for all t , the particles' common frame is the frame of the relative coordinate $x_A - x_B$, and “co-located” acquires a number, the rms $\sqrt{2}w_0e^{-r}$ of the relative coordinate, 5.2 nm for 10 nm packets at $r = 1$; the Reid ratio $1/\cosh^2 2r$ and Duan sum $2e^{-2r}$ are the same in all three frames. Under free flight the relative coordinate spreads as $\text{Var } x_{\text{rel}}(t) = 2w_0^2e^{-2r} + 2w_p^2e^{2r}t^2/m^2$, driven by the anti-squeezed $p_1 - p_2$: for electrons with $w_0 = 10 \text{ nm}$ and $r = 1$ the rms doubles in $t_2 = 4.05 \times 10^{-13} \text{ s}$, $e^{2r} = 7.39$ times faster than an unsqueezed pair, while a doubling time of 1 ns needs $w_0 = 0.497 \text{ }\mu\text{m}$. The entanglement itself does not decay — $\nu_- = (\hbar/2)e^{-2r}$ and $\text{Var}(p_1 \pm p_2)$ are constants of motion, and the back-evolved combination $x_1 - x_2 - (p_1 - p_2)t/m$ witnesses it at all times — but the fixed-quadrature Reid and Duan witnesses are lost after $9.81 \times 10^{-13} \text{ s}$ and $8.36 \times 10^{-13} \text{ s}$ (Appendix B). A spin projection on either particle leaves mean and covariance unchanged while the spin concurrence drops to 0, and CHSH is $2\sqrt{2}$ for every r . Massive-particle EPR correlations have been measured with metastable helium pairs from a condensate collision — created co-located and flying apart with $p + p'$ fixed, the geometry of this interpretation [120,121] — and with collective spin quadratures of an atomic ensemble [122]; no position–momentum EPR measurement on an electron pair was found. Every standard interpretation predicts the same covariance dynamics; only dynamical collapse would add a diffusive term (Sec. 3.9).

3.6 Many particles: the locality graph, GHZ states, entanglement swapping and mixed states

This subsection asks whether the two-particle construction extends to any number of particles, whether it survives the operations by which entanglement is created, redistributed and destroyed in the laboratory, and

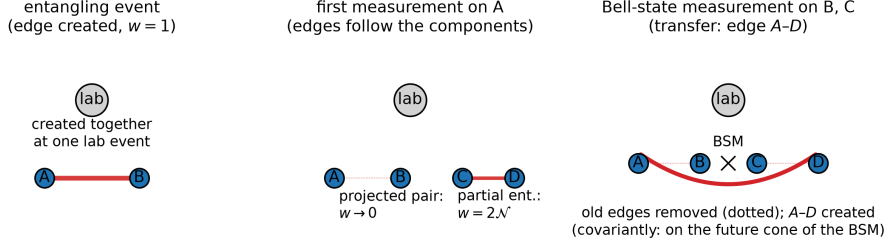


Figure 2: The locality graph at the three kinds of event (schematic; edge thickness is the weight w). Left: a co-located entangling event creates an edge. Middle: the first measurement on one particle splits its component and the edges follow (P2), with partial entanglement as a partial weight. Right: a Bell-state measurement on two partners removes their edges and creates the edge between the particles they were paired with (P3); in the covariant variant the change reaches each chart on the future light cone of the event (P4).

what it says for mixed states (Fig. 2). Every number below was computed numerically; the spin statistics are those of quantum mechanics throughout.

The locality-graph rule, and the label. Particle i has a height $z_i \neq 0$, a lab velocity v_i , and each pair carries a weight $w_{ij} \in [0, 1]$; the potential of particle i is pinned by P1 (Eq. (7)) and interpolated between the pinned heights by the lowest-degree polynomial. For $N = 2$, $z = \mp 1$, $v_{A,B} = \mp v$ and $w = 1$ the rule returns the potentials Eq. (3) to 10^{-12} ; every pinned velocity is exact on a five-particle example (two pairs and an unrelated fifth); $N = 3, 4, 5$ particles with distinct lab velocities and all-pairs edges are mutually at rest at N distinct heights, so a one-dimensional label encodes any graph. Because σ_{lab} is single-valued, two particles at the same height must be co-moving in the lab — a labeling constraint, since distinct heights are always available. And the label is only a label: every frame map is only ever evaluated at a pinned height, so only the pinned values of Eq. (7) enter any prediction; re-embedding the same pinned data in two label dimensions lowers the interpolant degree from N to about $\sqrt{2N}$ while every velocity of every particle and of the lab in every frame agrees with its one-dimensional value to 10^{-8} . The dimension of z is not a parameter of the model, and the variant in which the coordinate is a physical dimension is refuted on other grounds (Sec. 3.3).

Three particles: GHZ states. Three particles born at one lab event in $(|000\rangle + |111\rangle)/\sqrt{2}$ [123,124], at three distinct heights with distinct lab velocities, have potentials of degree 3 and sit at $x = 0$ in each other's frames. The Mermin value [125] is 4 against the local bound 2, the sign constraints being satisfied by none of the 64 deterministic local assignments. Every two-particle reduced state of GHZ is separable, so a graph weighted by pairwise concurrence would be edgeless for a jointly entangled triple: the graph must be defined by the entangled components of the state, and the per-edge rule of P2 does this — for the pure GHZ triple every cut **separating** two particles has negativity $1/2$, so all three edges carry weight 1 although every reduced pair is separable. With it, measuring particle 1 along Z leaves 2 and 3 in $|00\rangle$ or $|11\rangle$ and all three edges go, including the one between 2 and 3, which were not touched; measuring along X leaves $(|00\rangle \pm |11\rangle)/\sqrt{2}$ and the edge stays; at polar angle θ from the spin $\hat{z}$ axis, $C_{23} = |\sin \theta|$. A measurement on one particle can thus change the locality between two others far apart in the lab — a non-local graph update in the lab which, in the particles' frames, where all three are at one point, is a local event; a pair created by such **pruning** keeps, in the covariant variant, the defining event of the component it was pruned from as the anchor of its ordering clock (P4): the pruning measurement changes its charts, not its clock. The frame potentials are thus functions of the global state — the only way the construction encodes monogamy — a state-dependent coordinate choice (Sec. 1). For a white-noise mixture $\mathcal{V} |\text{GHZ}\rangle\langle\text{GHZ}| + (1 - \mathcal{V})\frac{1}{8}$ the Mermin value is $4\mathcal{V}$; the published GHZ data are the photonic Mermin tests of [126] (Mermin value 2.84 from the reported fraction 0.85 ± 0.04) and [127] (2.77 ± 0.08 , $\mathcal{V} = 0.69$) — consistency checks only — and the ion-trap GHZ states of [128] ($N = 4, 5, 6$, fidelity lower bounds 0.76(1), 0.60(2), 0.509(4)) and [129] ($N = 14$, fidelity 50.8(9)%, meeting the witness line by 0.9σ), each one component, charted between preparation and readout. All were created co-located, which is the construction's own creation postulate; it adds nothing to their statistics.

Entanglement swapping. Pairs $A \sim B$ and $C \sim D$ are created at separate events; B and C are brought together and measured in the Bell basis, after which A and D , which never met, are entangled. Four particles at heights $(-1, 1, -2, 2)$ with distinct lab velocities were run with the Bell-state measurement applied as a projection at the meeting event, with the outcome-dependent correction on D , in both time orders: before the measurements of A and D (case a) and after both (case b). Positions of all four particles are continuous through

the transfer event in all five frames. In A 's and D 's frames their separation is constant while the edge exists — at rest relative to each other though they never met, at constant separation rather than co-located — and grows before and after. Each Bell outcome has probability $\frac{1}{4}$; the conditional A – D state has fidelity 1 with a Bell state and, after the correction, $|S| = 2\sqrt{2}$ for every outcome; without the correction the outcome-averaged state is maximally mixed and $S = 0$, so nothing is signaled. The separable-state measurement used as a control by [130] leaves A – D in a product state with witness $W = \frac{1}{2} - F = 0$, against $W = -\frac{1}{2}$ for every Bell outcome — the sign pattern of [130], -0.181 ± 0.034 against $+0.078 \pm 0.029$; two imperfect sources of Werner weights p_{W1}, p_{W2} give a swapped pair of weight $p_{W1}p_{W2}$. The two time orders give the same joint distribution of the three outcomes to 10^{-16} , because the projectors commute. The construction's account differs. In case (a) the edge A – D exists between the transfer and A 's measurement, and the two-particle story holds for a pair that never met. In case (b) A 's measurement changes B and D 's changes C ; the Bell-state measurement then finds B and C with no partners and transfers nothing; A and D are never local at any time, and their correlations appear only when the records are sorted by the later outcome. The statistics are order-independent; the construction's narrative is order-dependent, and we state this plainly as a cost. Peres's analysis of delayed-choice swapping sharpens it: the A – D conditional state exists only given the Bell outcome in case (a) as well, so even there the narrative is bookkeeping relative to that outcome [131] (Egg argues that this poses no threat to realism about entanglement as a relation [132]; the cost here is to the narrative). The covariant variant adopts the retarded reading of the transfer postulate (P4): the edge A – D comes into being where the transfer's future light cone reaches each partner — empirically inert by Proposition 1(iii), verified to leave every statistic unchanged — which makes edge existence frame-independent (Sec. 3.7) but does not dissolve the delayed-choice cost: in case (b) the outer measurements lie outside the transfer's future cone in every frame. A variant without transfer is observationally identical; the transfer postulate is forced by the construction's definition of locality, not by data. Consistency with experiment: [133] reported a swapped-pair visibility 0.65 ± 0.02 ; [134] measured $S = 2.421 \pm 0.091$ with the fidelity unchanged when the Bell-state measurement was delayed until after the outer detections; in [130] the middle measurement came 485 ns after the outer ones; with independent sources [135] obtained $S = 2.37 \pm 0.09$ and 2.38 ± 0.09 ; no experiment reports a difference between swapped and directly created entanglement, and the construction predicts none. Deterministic swapping with four trapped ions, at a fidelity of about 79% [136], is a matter-only Bell-state measurement, exactly the operation P3 describes.

One gap in this record must be stated rather than papered over. The loophole-free spin Bell test of [4] entangled spins 1.3 km apart by heralded swapping at a midpoint — but the Bell-state measurement there is performed on **photons** emitted by the two spins, as it is in the event-ready atom test of [9], and the circuit test of [10] distributes its entanglement over a microwave-photon link. A spin–photon pair carries no edge to transfer, because photons carry no label, so the mechanism cannot say how those spins became edge-connected. We resolve this by declaring photon-mediated entangling steps outside the present scope: the graph of those experiments is defined only from the point at which the matter pair exists, the heralding itself is unmodeled state preparation, and the experiments enter our tables as consistency checks on the statistics alone. A provisional rule — reading the graph off the matter state after heralding — would cost an entangling step with no chart of its own, an audit that belongs to the mode-label extension (Sec. 4.3), not to this paper.

Mixed states. For a mixed state the finest product decomposition is undefined, and P2 states the rule directly on ρ , **per edge**: the weight of the edge (i, j) is the minimum, over the 2^{N-2} bipartite cuts that separate i from j (of the $2^{N-1} - 1$ bipartitions), of the negativity across the cut [110,111], normalized by $(d_{\min} - 1)/2$ with $d_{\min}$ the smaller of the two spin dimensions. The divisor is 1 for spin-1/2 and is what keeps $w \in [0, 1]$ for higher spins, where the bare doubled negativity reaches $d - 1$ (a maximally entangled spin-1 pair would otherwise get $w = 2$; with the normalization it gets $w = 1$, verified with $w \in [0, 1]$ on random two-qutrit and qubit–qutrit states) — a one-line theorem, since each singleton cut $\{i\}$ | rest is itself a separating cut with negativity at most $(d_i - 1)/2$. Components are the connected sets of the resulting graph; for a lone pair the rule reduces to $2\mathcal{N}(\rho)$; for $N \geq 3$ it is a definition, blind to bound entanglement [137].

Proposition 4 (continuity in the state). Every edge weight $w_{ij}(\rho)$, and with it every pinned potential value of Eq. (7), is a continuous function of the state. *Proof.* The negativity across a fixed cut is a continuous function of ρ (a sum of eigenvalues of a fixed linear image of ρ), w_{ij} is the minimum of finitely many such functions, and the divisor is a constant per edge. $\square$

Continuity is not automatic: the natural per-component alternative — the weight of a component taken as its weakest single-particle-cut negativity — fails it, since for $\sqrt{1 - \varepsilon^2}(\psi_{12}^- \otimes |0\rangle_3) + \varepsilon |111\rangle$ that rule gives the

1–2 weight $2\varepsilon \rightarrow 0$ as $\varepsilon \rightarrow 0$ although the pair’s marginal approaches a pure singlet, then jumps to 1 at $\varepsilon = 0$: a nearly decoupled spectator annihilates the locality of a singlet it barely touches. Under the per-edge rule the same family gives $w_{12} \rightarrow 1$ smoothly while $w_{13}, w_{23} \rightarrow 0$ as $\approx 2\varepsilon$ — verified along the whole family, with Proposition 4 under random perturbations. Thresholds were computed by bisection and compared with published values. For the noisy GHZ triple $p_W |\text{GHZ}\rangle\langle\text{GHZ}| + (1 - p_W)\frac{1}{8}$ each cut has negativity $\max(0, \frac{p_W}{2} - \frac{1-p_W}{8})$: the component exists for $p_W > \frac{1}{5}$, the exact separability point [138], and for general N for $p_W > \frac{1}{1+2^{N-1}}$, while the projector witness certifies genuine multipartite entanglement only for $p_W > \frac{3}{7}$ [139] — for general N above $\frac{2^{N-1}-1}{2^{N-1}}$, a ratio of 2.14, 4.2, 16.2 and 4096 to the component threshold at $N = 3, 4, 6, 14$ — and the Mermin value $4p_W$ exceeds 2 only for $p_W > \frac{1}{2}$. For the noisy W triple the cuts become positive below $p_W = 0.2096$, the pairwise concurrence dies at 0.548 and the projector witness $(2/3)\mathbb{1} - |W\rangle\langle W|$ certifies above $\frac{13}{21}$ (re-derived in one line from the witness): between 0.21 and 0.55 the component rule gives one component while a pairwise rule gives no edge. Under local dephasing at rate γ per qubit ($e^{-\gamma t}$ convention) the cut negativity of an N -qubit GHZ state is exactly $\frac{1}{2}e^{-N\gamma t}$, so the component persists with weight $e^{-N\gamma t}$; local depolarizing of strength q_{dep} instead ends it at $q_{\text{dep}}^* = 0.4226, 0.4433, 0.4226$ for $N = 2, 3, 4$, not monotone in N .

Two caveats close the subsection. The graph is read off a density operator, which can be a **proper** mixture (ignorance of a pure state) or an **improper** one (the reduced state of a larger entangled system); the rule treats both alike, so the ontic geometry is read off a possibly epistemic object — part of the interpretation’s price, since P2 names which state counts. And the cost is definitional: a noisy GHZ state stays one component down to $p_W \approx 2^{1-N}$ but loses certifiable genuine multipartite entanglement at $p_W \approx \frac{1}{2}$, so the construction’s locality outlives certifiable entanglement by a factor of about 2^{N-2} in p_W ; whether “one component” should mean negativity across every cut, as adopted, or genuine multipartite entanglement is a choice the data do not make.

3.7 A relativistic version

Everything above uses an absolute lab time t , and the rule that the earlier of two measurements projects the singlet needs a meaning for “earlier”; for spacelike-separated measurements — the Bell configuration — the lab order is frame-dependent. This subsection reports a covariant version, first for a pair created at one event and then for components defined at an event on neither partner’s worldline, restricted to $1+1$ dimensions and massive particles (in $1+1$ dimensions the little group is trivial, so “spin” is an inert internal label); every statement was verified numerically ($c = 1$).

3.7.1 Boost potentials and proper-time ordering

The shear is replaced by a Lorentz boost whose rapidity depends on the height: the map from the lab to frame k at height z is $(t', x') = \Lambda(\psi_k(z))(t, x)$ with $\Lambda(\psi) = \begin{pmatrix} \cosh \psi & -\sinh \psi \\ -\sinh \psi & \cosh \psi \end{pmatrix}$, and from frame k to frame l it is $\Lambda(\psi_l(z) - \psi_k(z))$; rapidities add, so composition holds identically. The pinning is the locality-graph rule with relativistic relative velocities: the pins at the particle’s own height and at the laboratory coincide, $\psi_k(z_k) = \psi_k(0) = \text{artanh } v_k$ (frame k is at rest in itself and sees the laboratory receding at $-v_k$), and

$$\psi_k(z_j) = \text{artanh } v_j - \text{artanh } u_{kj}, \quad u_{kj} = (1 - w_{kj}) \frac{v_j - v_k}{1 - v_j v_k}, \quad (8)$$

interpolated between the pinned heights as before, so particle j moves at u_{kj} in frame k : at rest for a partner, at the relativistic relative velocity for an unrelated particle. For a partner $\tanh \psi = v_j$, so the event $(t, v_j t)$ maps to $x' = 0$ and $t' = t\sqrt{1 - v_j^2} = \tau_j$: partners are at rest at $x' = 0$ in each other’s frame, and the frame time of a partner’s measurement event is that partner’s proper time since creation. The ordering rule becomes: the smaller proper time projects the singlet. In a working pair with $v_A = -0.3c$, $v_B = 0.8c$ (deliberately asymmetric), measured at $t_A = 1.0$, $t_B = 1.2$, $\tau_A = 0.95394$ and $\tau_B = 0.72$, so B is first although A is first in the lab; laboratories moving at $u = -0.5c, 0.2c$ and $0.9c$ return the same proper times to 10^{-12} and the same order. For $t_A < t_B$ the two orders disagree exactly when $1 < \frac{t_B}{t_A} < \frac{\gamma_B}{\gamma_A}$, and for measurement times uniform on the unit square the disagreeing fraction is $\frac{R-1}{2R}$ with $R = \frac{\gamma_{\text{max}}}{\gamma_{\text{min}}}$: 0.1855 for the working pair (Monte Carlo 0.181 ± 0.003), zero for symmetric velocities (Fig. 3). The disagreement is confined to spacelike separations: both proper times are intervals from the creation event, and by the reverse triangle inequality the later of two causally related events always carries the larger one (of 2×10^4 random configurations with velocities in $(-0.95c, 0.95c)$, every timelike or lightlike pair had the same proper-time and lab order). Where relativity defines an order the rule agrees with it; where relativity leaves it undefined the rule supplies an invariant one — a property the general

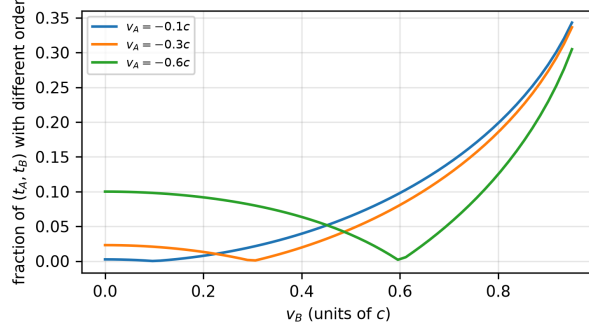


Figure 3: Fraction of measurement-time pairs (t_A, t_B) , uniform on the unit square, for which the proper-time order differs from the lab order, against v_B for three values of v_A ; the closed form is $\frac{R-1}{2R}$. Every disagreeing pair is spacelike separated. For the matter-system speeds of the in-scope experiments the fraction is $\approx 10^{-9}$ everywhere on this plot: the figure describes a regime those experiments never enter.

rule below keeps by construction. The rule is undefined on the tie set $\tau_A = \tau_B$, which contains the standard symmetric Bell configuration (equal speeds, simultaneous laboratory measurements); on it both orders produce identical statistics (Proposition 1(iv)), so the ordering is left undefined there — and with it the state history of Sec. 3.7.3, since which wing’s projection changed the partner is then undefined too. For the matter-system speeds the mechanism describes the tie set is not a line but a blur: the two orders differ only in a window of relative width $(v_B^2 - v_A^2)/2c^2 \approx 1.7 \times 10^{-9}$ for electrons at 10^4 and 2×10^4 m/s, and inside a sub-window of relative width $\approx 6 \times 10^{-11}$ the order is undetermined by the $\pm 1\%$ track-velocity uncertainty of Sec. 3.5 — the worldlines the clock is anchored to are conditional means with that noise. In the non-relativistic limit $v = \mp \varepsilon$ the boost potentials reproduce the shear potentials with relative error $0.1875\varepsilon^2$ in the norm $\max_z |\sigma^{\text{boost}} - \sigma^{\text{shear}}| / \max_z |\sigma^{\text{shear}}|$ over $z \in [-1, 1]$ (the maximum sits at $z = -1/2$); the pointwise relative deviation of the velocity field is $\varepsilon^2(1 - q^2)/3$ with $q = z^2 + z - 1$, approaching $\varepsilon^2/3$ near the potential’s zero at $z = (\sqrt{5} - 1)/2$. The spin statistics are unchanged in either order, $|S| = 2\sqrt{2}$ with signaling below 10^{-12} . The rule is not the multisimultaneity model of Suarez and Scarani [140], refuted by [141] (Sec. 3.8): it takes no device velocity as input.

3.7.2 Components defined off the worldlines: the interval clock

For a swapped pair the partners were created at different events, and proper time since creation is not a common clock: comparing the two proper times orders by the firing time of the second source, which changes no state, graph or statistic — invariant and physically arbitrary (anchoring the frame at A ’s creation gives frame times (5.8788, 5.1384) in the configuration below, D first; at D ’s, (5.5522, 6.0622), A first). P4 uses the component’s defining event $\mathcal{E}$ instead — the transfer $\mathcal{T}$ for a swapped pair, the creation or entangling event for a pair born together, and for a pair created by pruning the defining event of the component it was pruned from (the pruning measurement changes the charts through its cone, not the clock) — and gives every member the same clock, the signed invariant interval from $\mathcal{E}$ to the measurement event $e_i = (t_i, x_i)$:

$$\kappa_i = \begin{cases} +\sqrt{(t_i - t_{\mathcal{E}})^2 - (x_i - x_{\mathcal{E}})^2} & \text{if } e_i \text{ lies in the future cone of } \mathcal{E} \\ -\sqrt{|(t_i - t_{\mathcal{E}})^2 - (x_i - x_{\mathcal{E}})^2|} & \text{otherwise} \end{cases} \quad (9)$$

— the proper time of an inertial clock carried from $\mathcal{E}$ to e_i , with the sign recording whether the change defined at $\mathcal{E}$ has reached the event (the negative branch covers events spacelike to $\mathcal{E}$ and events in its past cone alike, only the sign being used; on the cone itself $\kappa = 0$); only $\mathcal{E}$ and the measurement event enter. For a pair created at $\mathcal{E}$ the worldlines pass through $\mathcal{E}$ and κ_i is the proper time since creation, so the rule above is the special case. Its level sets inside the future cone of $\mathcal{E}$ are the hyperboloids $t^2 - x^2 = \kappa^2$ — the Milne foliation of that cone, the natural simultaneity of the defining event: the ordering uses no foliation of spacetime, but it does use one of the component’s own light cone (Sec. 3.7.4).

Proposition 5 (invariant ordering and chart locality). (a) Under P4 the measurements of a component are ordered by a strict partial order that is Lorentz-invariant, agrees with the causal order wherever relativity defines one, and is total, up to ties, on the events inside the future cone of the defining event. (b) Under the retarded reading no frame potential changes outside the future light cone of the component-changing event that causes the change, in any frame; and the change of potentials alters no spin statistic. *Proof.* (a) κ is a function

of an invariant interval, hence invariant, and in the open cone it is strictly increasing along every future-directed causal curve: if e_1 precedes e_2 causally, the reverse triangle inequality gives $\kappa(e_2) \geq \kappa(e_1) + \tau(e_1 \rightarrow e_2)$, strict for a timelike step, and for a null step ($\tau = 0$) strictness follows from $\kappa(e_2)^2 = \kappa(e_1)^2 + 2(e_1 - \mathcal{E}) \cdot (e_2 - e_1)$, the Minkowski product of a future timelike and a future null vector being positive; on the cone's boundary $\kappa \equiv 0$ along a generator, and such pairs are ordered by the causal clause. The κ -order on in-cone events therefore contains the causal order there, and the relation P4 defines — the transitive closure of the causal order together with the κ -order on in-cone events — is acyclic: a chain that enters the cone never leaves it (every causal successor of an in-cone event is in the cone), inside the cone the relation is the κ -order, and among events outside it the causal order alone. An acyclic transitive relation is a strict partial order; on in-cone events it is total up to ties. (b) Particle i 's potentials change at the invariant event where the cone of the causing event meets i 's worldline, and cone membership is invariant; the second clause is Proposition 1(iii). $\square$

Clause (b) is a definition read back: it says that P4 is consistent and invariant, nothing more. Clause (a) is the theorem the variant needs, and one remark explains why the rule stops where it does: in the spacelike region outside $\mathcal{E}$'s causal cone no Lorentz-invariant clock can respect the causal order — every invariant level set there (the hyperbolas $x^2 - t^2 = s^2$) contains causally ordered pairs of events, for instance $(0, 5)$ and $(30, \sqrt{925})$ — so P4 orders those events causally and not otherwise (in $\mathcal{E}$'s past cone κ is itself monotone along causal curves; the delayed-choice events below lie there, and P4 orders them causally too); and under the retarded P2 the region outside the cone is exactly where the component's edges have not yet reached the charts, so the clock orders precisely the measurements the charts pair.

Verification. Pairs $A \sim B$ and $C \sim D$ were created at $(t, x) = (0, 0)$ and $(0, 1.6)$ with velocities $(-0.2, 0.4, -0.4, 0.5)c$; B and C meet at $\mathcal{T} = (2, 0.8)$; A and D are measured at $t_A = 6$, $t_D = 7$, spacelike separated, and, in a second run, at $t_A = 3$, $t_D = 4$, before the transfer's cone reaches either. In the first run $\kappa_A = 3.4641$ and $\kappa_D = 2.5515$, so D is first, invariantly ($u = -0.5c, 0.2c, 0.9c$, to 10^{-12}). In the early run $\kappa_A = -0.9798$ and $\kappa_D = -1.9596$: both measurements lie outside the cone of $\mathcal{T}$ — where, under the retarded reading, no A – D edge exists in either chart — and, being spacelike separated, they are unordered by P4; the sign of κ is itself invariant and marks cone membership, so edge existence is an invariant, verified over 400 random configurations and boosts together with the equivariance of the cone-crossing event at which each particle's chart changes. Proposition 5(a) was checked on 2×10^5 random two-particle configurations with an off-worldline defining event and both measurements inside its cone (the 4×10^4 timelike-separated pairs among them are all ordered causally, where a clock zeroed on each worldline's own cone crossing would invert 3.6% of them); on 1800 random components of two to four members with events inside and outside the cone (strict partial order, causal consistency and boost invariance in every case); on a GHZ triple measured at three mutually spacelike events (a total order by proper time from the creation event, invariant); and on the pair left by pruning such a triple, which keeps that order under the inherited anchor; anchored at the pruning measurement instead, the remnant pair of a mutually spacelike triple would be unordered, and inside that measurement's cone the two anchors can disagree (worldlines from the origin at 0, $-0.633c$, $0.200c$ measured at $(1, 0)$, $(3, -1.9)$, $(2.2, 0.44)$: $2.156 < 2.322$ from the origin, $0.624 < 1.116$ from the first measurement — opposite orders). For the delayed-choice geometry of [130], with massive stand-ins at rest at $x = \mp d$ measured at $t = 0$ and the transfer 485 ns later (the experiment's delay; its distance is not recorded, so d is a parameter), the clocks are -485.0 , -483.9 and -352.1 ns for $d = 1, 10$ and 100 m, identical under global boosts $\beta = 0, \pm 10^{-3}$: both outer particles were measured before any locality between them existed, in every frame, and the two measurements are unordered by the rule. The narrative gap of the delayed-choice ordering recorded in Sec. 3.6 is covariant, not a lab-frame artifact. The statistics are indifferent throughout: projecting A or D first gives the quantum box to 10^{-12} .

What “located” and “ordered” now cover. First, the ordering lives on events, not in charts. Chart k reads the graph off the state on its own past light cone (P1, P2), and in a spacelike Bell configuration each measurement event lies outside the other's future cone: at the κ -later event the partner's chart still shows the pair edge-connected and at rest, although the conditional state is already a product, and each chart records its own measurement as the one that split the pair. The κ -verdict is written into no chart, no past-cone state and no statistic; it is an invariant partial order on the two events and nothing more, and “ordered” in this paper means exactly that. Second, “located” is earned at the first (κ -earlier) measurement of a creation-anchored pair, where the acted-on factor and the passive partner sit at one chart point; at the second measurement the measuring wing's own chart is stale by a light-crossing time in every spacelike Bell test, and the symmetric configuration sits on the (blurred) tie set. Chart and state thus disagree inside the light-crossing gap in both directions — the edge outlives the entanglement it was read off for up to a light-crossing time, and a swapped

pair is entangled from the projection onward but not yet edge-connected outside the transfer's cone. The chart is read from the past-cone state; the κ -ordered history is event-level bookkeeping; the two are not rival state assignments but different objects, a graph and a clock.

Component orders do not compose. P4 orders the projections **within** a component. The component orders, each a strict invariant partial order, do not in general compose with one another and with the causal order into a partial order of all projections on S . With pairs created at $(0, 0)$ and $(0, 1)$, B and C meeting at the Bell-state measurement $\mathcal{T} = (1, 0.5)$, D measured at $(2, 2.8)$ and A at $(2.5, 2.35)$: A 's measurement is κ -before $\mathcal{T}$ in $\{A, B\}$ ($0.8529 < 0.8660$), $\mathcal{T}$ is κ -before D 's measurement in $\{C, D\}$ ($0.8660 < 0.8718$), and D 's measurement lies in A 's causal past — a cycle, which occurs in about 0.25% of random swapping geometries, always in the delayed-choice branch where an outer measurement precedes the Bell-state measurement in its component's order. Nothing statistical or chart-level depends on it (the instruments at unordered events commute, and charts are read off past-cone states); what it settles is that the state history of Sec. 3.7.3 is per component.

3.7.3 The state between the two measurements

The κ -clock orders the two projections invariantly; it does not say what the joint state **is** on a spacelike hypersurface Σ lying to the future of the smaller- κ measurement event and to the past of the other, in a frame where the coordinate order disagrees with the κ -order. That question has owned a literature for sixty years: Bloch's non-covariant wave function of a particle detected in spacelike-separated regions [142]; Hellwig and Kraus's reduction across the light cone [143]; Aharonov and Albert's argument that no event-anchored covariant reduction yields a consistent state history — the probabilities survive covariantly, the states do not — and their positive proposal of states as functionals of hypersurfaces [144–146], developed by Fleming into hyperplane-dependent reduction [147, 148] (Ghirardi reviews the constraints [149], Maudlin the terrain [150]); Myrvold's benign relativity of the state history [151]; and Peres's deflationary end of the same position, that states are not covariant but predictions are [152].

The variant's answer: the mechanism assigns states per component, to the invariant structure it uses — each component's partial order of events, each projection located at its own event and ordered by κ ; the reduced conditional state of a component is well defined because the instruments at events P4 leaves unordered act on different particles and commute (Proposition 1(iv)), so every linear extension of the component's order gives the same state, and no history of the joint state of S is assigned, since the component orders do not compose (Sec. 3.7.2) — and treats the interpolating hypersurface state as a chart artifact, as it treats the laboratory's simultaneity. Wherever P4 leaves two events of a component unordered — on the tie set, or both outside the defining event's cone, or one inside and one outside and spacelike — the reduced conditional state after both is defined, which of them changed the partner is not, and under iterated swapping neither is whether the earlier-measured member still belonged to the component the later transfer created. On a Σ between the two measurements, “projected or not” is hyperplane-relative bookkeeping in Myrvold's sense, or gauge in Peres's: nothing in the mechanism's ontology attaches to Σ , and no instrument of the class Proposition 1 admits evaluates a state there. That class must be stated precisely, because Aharonov, Albert and Vaidman showed that nonlocal joint observables of two separated particles can be measured by local interactions and detections [153] — instruments whose outcome does evaluate the joint state across a spacelike region. Proposition 1's instruments are each supported on one particle or on co-located particles (the Bell-state measurement of P3 is such a joint instrument, inside the class, with its own event); Aharonov–Albert nonlocal measurements lie outside the covariant variant: the κ -rule assigns them no event, and a mechanism whose ordering lives on events has nothing to say about an instrument with none. The position is adopted, not proven: whoever wants the state history itself to be ontic will find this variant as unsatisfying as every event-anchored covariant reduction since [143] — that is the Aharonov–Albert lesson, and we do not claim to have escaped it.

3.7.4 Scope and costs of the covariant variant

The scope of the covariance. The ordering half is dimension-blind. The interval clock of Eq. (9), the retarded charts of P4, Proposition 5 and the state-history position are statements about events, intervals and light-cone membership, and hold verbatim in any dimension: with the Euclidean norm in the spatial displacement the clocks, the cone sign and the order are invariant under random non-collinear boosts, and the disagreement fraction $(R - 1)/(2R)$ holds for perpendicular velocities (Appendix C). One caveat keeps this honest: in $1 + 1$ dimensions an invariant ordering of spacelike-separated events is cheap — the sign of Δx is itself boost-invariant — so it is the anchoring to a defining event, not the dimension, that carries the rule to $3 + 1$. The Galilean base form extends as cheaply, and composition is not among the things the construction must change: scalar

additive rapidities have no $3 + 1$ analog, but group-valued potentials $\Lambda_k(z) \in \text{SO}^+(1, 3)$, pinned by the same rule, give right-quotient frame maps $T_{kl} = \Lambda_l \Lambda_k^{-1}$ that compose and invert identically, with partners at $x' = 0$ and the collinear limit recovering Eq. (8) (Appendix C).

The genuine $3 + 1$ residue is in the spin sector, in three layers. First, maps between different particles' frames carry Wigner rotations — up to $\approx 27^\circ$ in the four-particle scan of Appendix C, a scan-specific number that does no work. On the sharp classical tracks the frames attach to, each particle has a definite velocity, so a boost acts on its spin by a **fixed** rotation — a local unitary per particle, which changes no cut negativity, hence no edge weight, and no laboratory statistic [154]; the general statement is not “the frame maps never touch the spin state”, a theorem of the one-dimensional restriction, but “they act by local unitaries, which preserve every cut negativity and every laboratory statistic”. Second, “partner at rest” fixes $\Lambda_k(z_j)$ only up to the rotation stabilizer of the partner's 4-velocity, so each pinned height carries an $\text{SO}(3)$ gauge; fixing it is a convention — the pure boost from the laboratory, the operational choice of [71] — but the transformation law for ρ under that convention has not been stated. Third, the genuinely open problem: for wave packets with momentum spread the spin transforms by momentum-dependent Wigner rotations, so the reduced **spin** density matrix is not covariant and its entanglement is frame-dependent [155–160]. P2's weight, read as a spin negativity, would then be frame-dependent in $3 + 1$ — incoherent for the rule that defines the charts. One candidate repair reads the cut negativity on the full one-particle factors (spin $\otimes$ momentum), with respect to which a Lorentz transformation is a local unitary on every particle cut; it is not a one-line change, because P2's normalization $d_{\min} - 1$ is then infinite and, unnormalized, the two-mode-squeezed motional pair of Sec. 3.5 — whose spin state is a product — acquires negativity $(e^{2r} - 1)/2 = 0.86, 3.19, 26.8$ at $r = 0.5, 1, 2$: a full-weight edge for a spin-separable pair, against the sector separation of Sec. 3.5. The existing solution to the relativistic Bell problem takes the other road: Streiter, Giacomini and Brukner define the spin in the particle's rest-frame quantum reference frame and obtain a maximal Bell violation without momentum post-selection [72] (the spacetime and clock versions of that program are [161,162]). Whether the intended weight is a spin $\otimes$ momentum negativity or a rest-frame spin negativity in their sense is a question this paper leaves open; the two give different graphs for the motional EPR pair. Either way one further cost stands: Proposition 1's hypothesis of a motional state that factorizes from the spin is not itself a Lorentz-invariant hypothesis. This residue is what avenue (iii) of Sec. 4.3 names (the Lorentz-QRF construction of [70] works in $1 + 1$ dimensions for a spinless particle, sharing the dimensional restriction, not the spin residue). A kinematic cost belongs beside these: the covariant chart applies the boost $\Lambda(\psi(z))$ from the origin, with no integrated analog of the Galilean form, so at a re-anchoring event chart positions jump — when the transfer's cone reaches A 's worldline in the working pair of Sec. 3.7.2, at $t = 3.5$, A 's chart moves its former partner B from 0 to 2.14 and its new partner D from 4.13 to 1.85 — and the continuity of positions stated for the Galilean charts (Sec. 1.1) does not hold in the covariant variant as built.

Photons. The variant excludes photons, and the reason is co-location, not ordering. A photon has no rest frame — the pinning $\tanh \psi = v_j$ has no solution at $v_j = c$ — so P1 can give a photon pair no shared rest frame and no chart co-locates a photon with its partner. Ordering is available: the affine parameter a photon's own objects supply, $x^\mu = \lambda k^\mu$, gives $\lambda = t/\omega$, a Lorentz scalar whose order never flips under boosts (0 of 181 in the scan of Appendix C). It is not P4's clock: it is the $m \rightarrow 0$ limit of the mass-rescaled clock $\kappa/m = t/E$, which P4 does not use — P4 compares κ itself, mass-independent, and the interval clock of Eq. (9) reads zero on every null worldline from the defining event, the tie set of Appendix C — and it depends on the photon's four-momentum, not on the event alone. For photons the covariant variant can therefore deliver “ordered” but never “located”, and it is the locating that the charts exist for; the mode-label extension of Sec. 4.3 inherits the same split (Sec. 3.8 states the consequence for the moving-apparatus experiments).

Costs, and foliations. The covariant version buys an invariant ordering and an invariant graph, at the priced cost that chart and state disagree inside the gap; it still pays with a fourth postulate, chosen for invariance among empirically identical rules, with $1 + 1$ dimensions and no gravity, with the exclusion of photons, and with a rule — projection ordered by the signed interval from the component's defining event — that belongs to no established relativistic quantum theory: the version is compatible with special relativity, not embedded in a field theory. Describing it as trading a preferred foliation for none needs qualifying: the interval clock foliates the future cone of each component's defining event by Milne hyperboloids — an event-determined foliation, covariantly fixed by the component's own history (“a faster particle measures later at equal laboratory times” says that the laboratory slice is not a Milne slice). The comparison with relativistic Bohmian mechanics is therefore on equal terms: Dürr, Goldstein, Norsen, Struyve and Zanghì argue that a foliation determined

covariantly by the wave function installs no additional Lorentz-violating structure and that the resulting theory can qualify as fundamentally Lorentz invariant [29], and Tumulka’s flash-ontology GRW dynamics needs no foliation at all [57] (Maudlin surveys the options [150]); the present foliation is determined by events rather than by the wave function, and both proposals avoid **extra** non-covariant structure. The claim is thus narrower than “no foliation”: every frame computes the same κ ’s from the same events, so the ontology carries no frame-dependent element, and the rule is the only clock the mechanism’s own objects supply. If nothing statistical depends on the ordering, why carry it? Because it gives “the first projective measurement changes its partner’s conditional state” a frame-independent referent within each component (Sec. 3.7.3), and that referent is carried by the events and by nothing else. Among collapse models the rule is nearest in spirit to the past-light-cone construction of [58], but unlike it changes no statistic.

3.8 Consistency with moving-apparatus and speed-of-influence experiments

Experiments have moved the measuring devices so that each wing is “first” in its own rest frame, and have timed distant measurements closely enough to bound the speed v_{inf} of any influence between them (Table 3). The mechanism’s prediction follows from one property of the frame maps: every map (Eq. (2)) has $t' = t$, so the order $\text{sgn}(t_B - t_A)$ of two measurement events is the same in the lab, in either electron’s frame, and in every frame built from the same potentials (confirmed for 200 random event pairs, speeds and heights); a laboratory moving at $-u$ along x is a new frame with potential $\sigma_{\text{lab}}(z) = vz + u$, which adds the same u to every map, leaves β_{AB} unchanged, keeps the separation rate $2v$, and leaves the sequential-projection statistics identical ($|S| = 2\sqrt{2}$ and remote-setting-independent marginals to 10^{-12} for u up to $2000v$, closure of the new frame with the others to 10^{-14}). The order of projection is invisible in the statistics, which is why the model can have an absolute order without a signal; the apparent influence speed the lab infers, $v(t_A + t_B)/|t_B - t_A|$, is a lab-frame artifact, unbounded as $t_A \rightarrow t_B$, while in the particles’ frames $\Delta x = 0$ and no speed is defined. The model therefore sits above every bound in Table 3 for a structural reason, not by a margin — a null test that every standard interpretation passes too, and a refutation only of finite-speed models, themselves excluded on independent grounds since any finite-speed influence reproducing the quantum correlations would permit superluminal signaling [51]. The multisimultaneity model of Suarez and Scarani [140], which orders the two measurements by each **beam splitter’s** rest frame, was refuted by [141]; the present model is not that model — its order is a property of the lab time in the Galilean version and of the particle worldlines and events in the covariant version, never of a device — and the covariant variant’s one moving-apparatus prediction is that a massive-particle before-before experiment shows the full correlations for every detector velocity, because the clock of P4 depends on the worldlines and the defining event alone.

The limitation is one of scope, and it should be stated in one sentence: **neither variant of the mechanism describes any experiment in Table 3** — the Galilean form lacks the Lorentz kinematics the configurations exploit (its absolute t has no frame in which the order can flip, so it cannot address the $\beta = 0.9$ figure of [53]), and the covariant form cannot locate photons (Sec. 3.7.4), which every entry uses. What survives is the consistency argument, and it is valid: both variants predict the standard statistics and hence no motion dependence, which is what was observed.

3.9 The one empirically distinct rival: dynamical collapse

Of the accounts surveyed in Sec. 2, all but one reproduce the predictions of quantum mechanics exactly, and the comparison with them can only be made on non-empirical grounds. The exception is dynamical collapse:

experiment	configuration	$\Delta x, \Delta t$	published bound
Zbinden et al. 2001 [163]	absorber 105 m/s; before-before; visibility 83%	10.6 km, 5 ps	$0.7 \times 10^7 c$ (quoted as $10^7 c$ in [163])
Stefanov et al. 2002 [141]	beam splitters 2500 m/s; all orderings; visibility $85 \pm 5\%$	55 m, —	$4.6 \times 10^5 c$ (lab frame)
Salart et al. 2008 [52]	midpoint source; timing; visibility $87.6 \pm 1.1\%$	18.0 km, 323 ps	$> 10^4 c$ ($\beta \leq 10^{-3}$); $\rightarrow 1.8 \times 10^5 c$
Yin et al. 2013 [53]	free space; loopholes closed; 12 h	15.3 km, 350 ps	$\geq 1.38 \times 10^4 c$ ($\beta = 10^{-3}$)
Cocciaro et al. 2011 [164]	tabletop; frame speed $< 0.1c$	—	$> 0.6 \times 10^4 c$
Cocciaro et al. 2018 [54]	$\beta \approx 10^{-3}$; $\chi = \pi/2$	1.2 km, —	$> 5 \times 10^6 c$

Table 3: Moving-apparatus and speed-of-influence experiments and the published lower bounds on a finite influence speed.

This interpretation predicts no motion dependence and $v_{\text{inf}} = \infty$, above every bound.

the GRW model [13], its continuous form CSL [14,30] and the gravity-based Diósi–Penrose model [31,32] add a stochastic term to the Schrödinger equation, with three unavoidable side effects — momentum diffusion, heating and spontaneous radiation [15]. The mechanism’s only non-unitary step is projection at a measurement; between measurements the evolution is unitary, so it predicts exactly zero for all three, for every particle at every time — as every no-collapse interpretation does, so the experiment discriminates the **class**, not the mechanism. Two structural facts shape the test, both verified numerically on the CSL rates and geometry factors. Free electrons are blind to CSL — even at the enhanced rate $\lambda \approx 10^{-8} \text{ s}^{-1}$ implied by Adler’s latent-image argument [165], the momentum spread of a 10^4 m/s electron needs 1.6×10^3 years to reach 10^{-3} of its momentum (in the diffusion convention $D_p = \lambda(m/m_0)^2 \hbar^2 / 2r_C^2$; convention-dependent at order unity) — and the mass density is spin-blind, so the collapse generator commutes with every spin projection and no Bell test separates the two (a 16-dimensional two-spin/two-site model confirms $S = 2\sqrt{2}$ through physical and exaggerated dephasing alike). The sensitivity lives in macroscopic test masses: at $r_C = 10^{-7} \text{ m}$ the tightest mechanical bound is $\lambda < 2.0 \times 10^{-10} \text{ s}^{-1}$, from the multilayer cantilever in the Hz–kHz band [55,166] (a figure-read value), with stronger bounds from X-ray searches only under a white-noise assumption that a spectral cut-off above a kilohertz evades [167]; the GRW value 10^{-16} s^{-1} lies far below every bound, and the mechanism predicts zero Diósi–Penrose radiation as well [168]. A force-noise measurement in the band no such cut-off can evade, with a natural next-generation target of order $\lambda = 10^{-12} \text{ s}^{-1}$ at 100 Hz — roughly two orders of magnitude below the published bound — is the one further avenue that can discriminate the mechanism’s class from a rival (Sec. 4); its design is not this paper’s subject. A detection of force noise with the CSL spectrum and mass-density dependence refutes the mechanism together with every no-collapse account; a null result shrinks the surviving collapse region (it never closes, since $\lambda \rightarrow 0$ recovers quantum mechanics) and leaves the interpretation question exactly where Sec. 3 leaves it.

4 Discussion

4.1 What is proven and what it costs

Consistency and exclusion. Two registers, kept apart throughout: **charted between preparation and readout** are the co-located matter-system tests — [7], [8], [169] and the ion-trap GHZ states [128,129]; **consistent by inheritance** is the rest of the record examined (photonic Bell violations, photon-mediated matter tests, no-signaling and speed-of-influence experiments, long-storage entanglement, swapping with and without delayed choice, photonic GHZ correlations, every non-interferometric collapse null), where “consistent” means only that quantum mechanics is right there and the mechanism is quantum mechanics where it applies. Neither register is an achievement of the geometry: both follow from Proposition 1. Three readings of the hypothesis are excluded by data already in hand, each in the register stated in Sec. 3.3: the contact-rule class by its generic members, surviving only in the fine-tuned member that copies the quantum conditionals; the persistent influence by the no-communication theorem; the physical extra dimension, for its universal readings, by gravity and collider data and, for free motion, by the mismatch between microsecond spreading and hours of stored entanglement.

What the geometry delivers, stated once. Proposition 1 makes every consistency result near-tautological, so we state in one place what the mechanism adds beyond bookkeeping and what each item costs; Table 4 keys each claim to its support. The update is **located**: at the onset of the first measurement of a pair born together the acted-on tensor factor and the passive partner sit at one chart point — earned at that event, and there only; at the second measurement of every spacelike Bell test the measuring wing’s own chart is stale by a light-crossing time (Sec. 3.7.2), and swapped partners are at constant separation, not co-located. It is **chart-local**: no chart changes outside the future light cone of the event that causes the change (Proposition 5(b)) — a definition read back, which says that P4 is consistent and invariant and constrains nothing further. It is **ordered on events**: P4 orders a component’s projections by a strict, Lorentz-invariant partial order agreeing with the causal order (Proposition 5(a)), anchored to the component’s defining event with no device and no foliation of spacetime, at the price of an event-determined foliation of the component’s own light cone (Sec. 3.7.4); the order is carried by the events and written into no chart, no past-cone state and no statistic, and it is undefined on a tie set that the symmetric Bell configuration occupies — blurred, for matter-system speeds, to a relative width $\approx 10^{-9}$ in the measurement times. It is **extensible** to many particles, swapping and mixed states, and could have failed to extend (Sec. 3.6). And it **prices itself**: under delayed choice the locality narrative has nothing to do, in both time orders once Peres’s point is taken (Sec. 3.6); a bound-entangled pair — entangled but positive under partial transposition [137] — gets $w = 0$ from P2 and is entangled yet

fully **non**-local in the chart, so on the negativity rule “entanglement as locality” fails extensionally there; the charts are state-dependent (Sec. 1); and the correlation strength is imported. That last cost is the theorem at the center of the paper: co-location admits every one-way-no-signaling box (Proposition 2); the one axiom the geometry can be given — order-independence, itself an input from quantum mechanics and experiment — cuts that set exactly to the no-signaling polytope and stops (Proposition 3); the Tsirelson bound is not geometric. None of this says **why** the correlations have quantum strength. “Valid interpretation” is claimed in exactly this sense: the structural claims are consistent with all verified data, the costs are stated, and the one prediction — collapse noise exactly zero, shared with every no-collapse account — is testable.

One-wing implementation, and what a contact-point jump law can be. The natural next demand is a local generator of the update in the particle charts whose laboratory pushforward is projection. The joint update of a singlet at the first measurement is effected by $P_s \otimes 1$, supported on the measured wing’s tensor factor alone — on the singlet, $(P_s \otimes 1)\rho(P_s \otimes 1) = (P_s \otimes P_{-s})\rho(P_s \otimes P_{-s})$, one line of algebra, and not ours: an instrument acting on one tensor factor while the distant “change” is conditional-state bookkeeping is the standard operational reading of projection, the engine of every no-communication proof [1] and explicit in the QBist literature [34], and it holds in **every** chart, the laboratory’s included, where the particles are $2vt$ apart — support on a tensor factor, not in space. What the label chart adds is exactly one thing: at the measurement event the acted-on factor and the passive partner sit at the same chart point. The two-sided identity covers maximally entangled pairs only (for $\cos\theta |01\rangle - \sin\theta |10\rangle$ the partner is steered into a non-orthogonal conditional state), so on the partial-entanglement sector only the one-wing reading survives: “locality of implementation” is earned in that qualified sense alone. What a jump law at the contact point can be is then fixed by the box it must reproduce — Proposition 2 read backwards:

Proposition 6 (the averaged contact conditionals are the quantum ones). Any outcome-producing rule supported at the contact point whose laboratory pushforward reproduces the singlet box has, averaged over whatever it carries from the source with the posterior weight given the first wing’s setting and outcome, exactly the quantum conditional probabilities. *Proof.* By the chain rule: for any rule, deterministic or stochastic, with $P_A(a | s, \lambda)$ and $P_B(b | s, a, s', \lambda)$ and λ independent of the settings, the posterior-averaged conditional $\sum_\lambda \mu(\lambda | s, a) P_B(b | s, a, s', \lambda)$ equals $P(a, b | s, s')/P_A(a | s)$ on the support of P_A , which is the quantum conditional whenever the box is the quantum one. $\square$

Contact rules with a shared variable λ fixed at creation do exist and are not thereby excluded: every no-signaling box, the singlet box included, is a mixture of the 24 deterministic contact vertices of [93] (the PR vertex is $a = \lambda$, $b = \lambda \oplus ss'$), so a deterministic λ -dependent jump law reproduces the singlet box without itself being the quantum conditionals — only its λ -average is. Such rules are parameter-dependent at the level of λ — as every deterministic Bell-violating model must be by Jarrett’s decomposition [87], a deterministic rule being outcome-independent — and no-signaling only on average, Shimony’s uncontrollable nonlocality [88]; the λ -free quantum-tuned rule of Sec. 3.2 is the outcome-dependent, parameter-independent member. Either way the geometry relocates Bell’s residue to one chart point and removes none of it. Locality of **explanation** is not available from the geometry. A remark closes the comparison with quantum reference frames: the chart velocity of a partner, $(1 - 2\mathcal{N}(\rho)) v_{\text{rel}}$, is a non-affine functional of the state (it has a kink at $p_W = 1/3$ along the Werner family), whereas a fixed unitary or channel read through a fixed observable induces affine functionals of ρ , so the chart velocity is not the expectation value of any fixed observable in any state-independently transformed state — which excludes only affine readouts of first-generation QRF transformations [67], a class to which no QRF quantity of interest belongs [73]. The honest comparison is the covariance-matrix one of Sec. 2.3.

The measurement problem: the horn taken and its price. The mechanism keeps single outcomes and the completeness of the state description; what it gives up is universal unitary dynamics, through a primitive projection triggered at “the first measurement” — with “measurement” undefined, exactly the primitive that Bell’s critique [20] charges against the orthodox account. On this axis the mechanism is Copenhagen-class, and its rivals that **solve** the problem — Bohmian mechanics with definite positions, Everett with universal unitarity, collapse models with a physical reduction — buy their ontologies with that solution, which the mechanism does not. Two structural questions are settled as stipulations, and priced as such. P2 names the reduced state of the labeled set S — the microscopic particles that carry labels — with the split between S and everything else a primitive of the same kind as the Heisenberg cut: read off a universal pure state, components would balloon into every apparatus and gas molecule the particles have ever scattered. During a measurement the apparatus enters as environment, its entanglement with S decoheres the reduced state of S , and for a single-particle measurement the edge weights fall continuously to the projected values (Proposition 4), so the charts change continuously

	claim	where	support
Prop. 1	empirical equivalence: distinguishable massive particles, instruments local to one or co-located particles, not coupling spin to position	Sec. 3.1	exact proof (near-tautological by design)
Prop. 2	contact rules = exactly the one-way-no-signaling polytope	Sec. 3.2	exact proof + enumeration
Prop. 3	order-independence (an imported axiom) narrows exactly to no-signaling, then stops	Sec. 3.2	exact proof
Prop. 4	every edge weight and pinned potential is continuous in the state	Sec. 3.6	exact proof
Prop. 5(a)	P4’s order is a strict, invariant partial order agreeing with the causal order	Sec. 3.7.2	exact proof, per component; checked on creation and transfer anchors, pruned components under the inherited anchor, timelike and spacelike pairs, in- and out-of-cone events, GHZ triples
Prop. 5(b)	no chart changes outside the future light cone of its cause	Sec. 3.7.2	definition + numerical check of cone-membership invariance (400 configurations)
Prop. 6	any contact-point jump law reproducing the singlet box has the quantum conditionals on average	Sec. 4.1	exact (chain rule); λ -contact rules exist

Table 4: The propositions: the claim, where it is established, and the kind of support. Definitional choices (the P2 weight rule, the P4 anchor, the split of S) are postulates, not propositions, and are priced here and in Sec. 3.7.

while the projection changes the conditional state; a Bell-basis projection on two partners sets the transferred edge discontinuously, the charts being read off the outcome-conditioned state (the canonical box’s “When”; Table 1 reads accordingly). For the in-scope experiments the stipulation is: in [7] S is the two ions’ spins, with the motional modes, the readout photons and the detectors outside it; in [8] S is the two electron spins, with the charge sensor and the reservoirs outside it, the spin-to-charge conversion of Pauli-blockade readout being the coupling at which Proposition 1’s instrument class ends. Whether the pointer is inside S would change the graph; the answer is stipulated, not derived, and “everything macroscopic sits at $z = 0$ ” is a statement about which systems carry labels, not a dynamical claim that measurement chains never entangle.

4.2 Where the mechanism stands among interpretations

Against Copenhagen the statistics are identical; the mechanism answers the question the orthodox account declines — where a separated pair is (at one chart point at the onset of the first measurement of a pair born together, in the state-dependent sense of Sec. 1) and why the laboratory reports the update as non-local — and shares Copenhagen’s unpaid debt on the measurement problem. Against Everett the mechanism keeps single outcomes by keeping projection, and pays with an extra coordinate where Everett pays with a universal wave function. De Broglie–Bohm is the closest in structure: both keep single outcomes and definite positions, and the Galilean form shares with Bohmian mechanics an absolute time in which the first measurement of a pair is an event at one instant; Bohmian mechanics has a fully specified dynamics, which the mechanism lacks, and the covariant variant’s event-determined foliation of each component’s light cone stands beside the wave-function-determined foliation of relativistic Bohmian mechanics [29] on equal terms — both avoid extra non-covariant structure (Sec. 3.7.4). Relational quantum mechanics and QBism relativize facts to systems or agents, not to coordinates; consistent histories and the transactional account are untouched by the mechanism’s postulates. Against ER=EPR the mechanism is a non-gravitational, tabletop-scale relative: the shear geometry plays the role of the bridge, no-signaling holds in both because nothing traverses the connection, and Proposition 2’s caution applies to both. Against dynamical collapse, and only there, the mechanism differs empirically, predicting no anomalous heating, force noise or radiation; every current null result is consistent with that prediction.

4.3 Further avenues

Five directions would change the picture rather than decorate it; none would make the mechanism the correct interpretation, and the first could show it to be wrong. (i) **A force-noise measurement** in the band no spectral cut-off above a kilohertz can evade, with a natural next-generation target of order 10^{-12} s^{-1} for the CSL rate (Sec. 3.9) — the only measurement that can discriminate the mechanism’s class from a rival; its design is outside this paper. (ii) **A Bell test in the mechanism’s own geometry** — two massive particles created together, separating, and measured by a spin-resolved detector each — is realized by a Cooper-pair splitter

[170–172]; the split electrons are identical fermions outside Proposition 1, so such a test would test quantum mechanics and the mode-label extension (v) under an arm-label reading it would first have to earn; it is not a proposal of this paper. For the record: the spin cross-correlation -0.96 reported on the InSbAs platform [173] is a conductance-based singlet–triplet indicator along one field axis, not a rotatable spin correlator; $S = 2\sqrt{2} \times 0.96 = 2.715$ would follow only if that visibility held at four freely chosen settings, a five-standard-deviation violation would then need 422 coincidences, none is possible below a symmetric filter polarization of 0.841 (the nanowire filters of [172], at ≈ 0.60 , give $S = 1.02$), and single-electron coincidence detection with two filter axes per side — which no splitter experiment has — would be the experiment, not an accessory to it. (iii) **A relativistic graph dynamics beyond 1 + 1 dimensions.** Composition is not the obstacle (Appendix C); the spin sector is — the wave-packet Wigner problem, the $SO(3)$ pinning gauge, and the choice between a spin $\otimes$ momentum negativity and a rest-frame-QRF spin negativity, for which the relativistic Bell test of Streiter, Giacomini and Brukner [72] is the existing solution and the competitor (Sec. 3.7.4) — together with an integrated covariant chart, gravity, and photons. (iv) **Mixed multipartite states beyond the partial-transpose rule**, where bound entanglement makes the component structure ambiguous: negativity is exactly what fails there ($w = 0$), and any continuous replacement keeps Proposition 4 while losing the $\ell = \mathcal{C}$ coincidences of Sec. 3.4. (v) **Labels on modes rather than particles** — on which the most waits: photons, identical particles, the splitter geometry of (ii). What transfers: P2’s negativity across a fixed mode bipartition, the min-cut, Proposition 4’s continuity, the $[0, 1]$ normalization (fermionic modes have $d = 2$), and, for null modes, an invariant ordering by the affine parameter $\lambda = t/\omega$ of Sec. 3.7.4 — which uses the mode’s four-momentum, not the event alone, and is sharp only for sharp frequencies — so an ordering, though not P4’s event-carried one, transfers. What does not: a mode is not a worldline, and a propagating photonic mode has no rest frame, so co-location, which the charts exist for, does not transfer; the mode basis is a new primitive (mode entanglement is not invariant under passive linear optics, and edges would join mode **groups**, which needs a coarse-graining rule P2 lacks); for fermions and photon number, superselection makes negativity overcount entanglement no local operation can use [174–176]; a single electron over two arms, $|1, 0\rangle + |0, 1\rangle$, has mode negativity $1/2$, so P2 read on modes would co-locate a single particle’s spatial superposition, which Sec. 1 disclaims; and polarization entanglement of beams of finite angular spread is itself frame-dependent [157,177]. The negativity computation and the ordering transfer; the rule does not, and co-location cannot.

Reproducibility. Table 5 maps the propositions and the principal numerical claims to the modules of the released package that verify them.

claim	section	module (tests in <code>tests/test_entangled_rNN.py</code> ; <code>graph_frames</code> has its own file, <code>lorentz_frames</code> is exercised under <code>r14</code> and <code>r16</code>)
Prop. 1, frame maps, CHSH invariance, tracking, EPR positions	Sec. 3.1, Sec. 3.5	<code>r01_baseline</code> , <code>r07_realistic_tracking</code> , <code>r08_epr_positions</code>
Props. 2, 3, 6; the 1024/128/256 enumeration; PR box as contact rule	Sec. 3.2, Sec. 4.1	<code>r02_contact_gate</code> , <code>r23_anti_distance</code>
persistent influence, E_2 , signaling 0.5, $N(F)$; physical dimension	Sec. 3.3	<code>r06_qd_at_projection</code> , <code>r04_physical_qd</code> , <code>r15_qd_dimension</code>
$\ell = w$, the two families, $2\mathcal{N} \leq \mathcal{C}$; graph, GHZ, swapping, thresholds, Prop. 4, $w \in [0, 1]$	Sec. 3.4, Sec. 3.6	<code>r05_continuous_qd</code> , <code>graph_frames</code> , <code>r11_locality_graph</code> , <code>r12_swapping</code> , <code>r13_ghz</code> , <code>r17_mixed_multipartite</code> , <code>r21_edge_weights</code>
boost potentials, proper-time order, $(R - 1)/2R$, NR limit; interval clock, Prop. 5(a), delayed-choice clocks; retarded updates, Prop. 5(b)	Sec. 3.7	<code>lorentz_frames</code> , <code>r14_lorentz_frames</code> , <code>r25_ordering_rule</code> , <code>r16_covariant_swapping</code> , <code>r22_retarded_updates</code>
3 + 1 charts, Wigner scan, photon clocks, singleton-cut bound	Appendix C	<code>r24_dim3</code>
moving-apparatus invariance; collapse rates and blindness; splitter numbers of (ii)	Sec. 3.8, Sec. 3.9, Sec. 4.3	<code>r03_absolute_time</code> , <code>r09_vs_collapse_models</code> , <code>r18_cps_proposal</code>

Table 5: Claim-to-test map for the released code (module names in the package’s `mapping_spaces/entangled/` directory; each numbered module has the test file of its number).

Appendix A The tracking model

Each electron's motional state is a Gaussian wave packet under continuous weak measurement of x with strength Γ_m (units $\text{m}^{-2}\text{s}^{-1}$), in the conventions of [114,115]: the stochastic master equation

$$d\rho = -\frac{i}{\hbar}[H, \rho] dt - \Gamma_m[x, [x, \rho]] dt + \sqrt{2\Gamma_m}(x\rho + \rho x - 2\langle x \rangle \rho) dW, \quad dy = \langle x \rangle dt + \frac{dW}{\sqrt{8\Gamma_m}}, \quad (10)$$

with $H = p^2/2m$ and record y . The conditional means obey $d\langle x \rangle = (\langle p \rangle/m) dt + \sqrt{8\Gamma_m}V_x dW$, $d\langle p \rangle = \sqrt{8\Gamma_m}K_{xp} dW$, and the covariances ($V_x, V_p, K_{xp} = \text{Cov}(x, p)$), which are deterministic, $\dot{V}_x = 2K_{xp}/m - 8\Gamma_m V_x^2$, $\dot{V}_p = 2\hbar^2\Gamma_m - 8\Gamma_m K_{xp}^2$, $\dot{K}_{xp} = V_p/m - 8\Gamma_m V_x K_{xp}$. Setting the derivatives to zero gives the minimum-uncertainty steady state $V_x = (\hbar/8m\Gamma_m)^{1/2}$, $K_{xp} = \hbar/2$, $V_p = \hbar(2\hbar m\Gamma_m)^{1/2}$, $V_x V_p - K_{xp}^2 = \hbar^2/4$, and, writing $D = V_x V_p - K_{xp}^2$,

$$\dot{D} = -8\Gamma_m V_x (D - \hbar^2/4), \quad (11)$$

so $D - \hbar^2/4$ decays monotonically and never changes sign: the Heisenberg product cannot be broken, and a mixed packet purifies (confirmed by integration from a mixed state and along a stochastic run). By the Itô rule the second moment of momentum grows as $2\hbar^2\Gamma_m t$, so the monitoring heats each electron at $\hbar^2\Gamma_m/m$: at $v = 10^4$ m/s the 0.28 meV kinetic energy doubles after 0.37 ms, 3.7 μs and 37 ns for $\Gamma_m = 10^{19}, 10^{21}, 10^{23} \text{m}^{-2}\text{s}^{-1}$, with steady-state widths of 1.10 μm , 347 nm and 110 nm. Fitting a straight line to the tracked separation over a run of length T at steady state gives

$$\text{Var}(\hat{2v}) = (2\hbar/m)(1.2/T + \gamma/2) + (52/35)\hbar^2\Gamma_m T/m^2, \quad \gamma = (8\Gamma_m \hbar/m)^{1/2}, \quad (12)$$

from the jitter of the means, the T -independent correlated position–momentum kicks, and momentum diffusion — the coefficients are the closed-form stochastic integrals $144 \int h^2$, $288 \int gh$ and $144 \int g^2$ of the least-squares kernel, with $\int g^2 = 13/5040$ (ensemble check at $\Gamma_m = 10^{19}$, $T = 100$ ns: rms 184.3 m/s against 183.9 m/s closed form). The relative error of $2v$ for an electron at 10^4 m/s is 0.92% at $\Gamma_m = 10^{19}$ and $T = 100$ ns, 7.3% at 10^{21} and 71% at 10^{23} ; the optimum run time is $T^* = \sqrt{21m/13\hbar\Gamma_m}$ and the optimal error degrades as $\Gamma_m^{1/4}$ (full table and figure in the released code). Pushing the tracks through the frame maps returns them exactly in the lab and, in A's frame, a tracked B with zero-mean separation and variance

$$\text{Var}[x_B - x_A]_{\text{frame A}} = 2 \left[\frac{\hbar}{m}t + \frac{8\Gamma_m V_x K_{xp}}{m}t^2 + \frac{2\hbar^2\Gamma_m}{3m^2}t^3 \right], \quad (13)$$

17.7 μm at $t = 100$ ns and $\Gamma_m = 10^{19}$ (ensemble: mean -0.02σ , rms 17.8 μm) against a lab separation of 2.0 mm.

Appendix B Free flight of an EPR pair

For the two-mode squeezed electron pair of Sec. 3.5 with $w_0 = 10$ nm and $r = 1$, free flight $V(t) = M_t V M_t^\top$ with $M_t : x_i \leftarrow x_i + p_i t/m$ gives, at $t = 0, t_2/2, t_2, 2t_2$ and $4t_2$ ($t_2 = 4.05 \times 10^{-13}$ s), an rms relative coordinate of 5.203, 6.882, 10.405, 18.758 and 36.418 nm, a Duan sum of 0.2707, 0.3722, 0.6767, 1.8947 and 6.7668 (lost above 2), a Reid ratio of 0.0707, 0.1220, 0.2682, 0.7560 and 1.8719 (lost above 1), and a constant partial-transpose eigenvalue $\nu_-/(\hbar/2) = 0.1353$. A helium-atom pair with $w_0 = 1$ μm has $t_2 = 2.95 \times 10^{-5}$ s. The tabulated values are in the released code.

Appendix C The charts in 3 + 1 dimensions

This appendix records, at the level of statements and numerical verification, the 3 + 1-dimensional form of the charts that Sec. 3.7.4 appeals to. It is a kinematic check, not the construction itself: the open spin-sector items of Sec. 3.7.4 are untouched here.

Galilean form. Give each particle a laboratory velocity $\mathbf{v}_i \in \mathbb{R}^3$ and let the potentials be vector-valued, $\sigma_k(z) \in \mathbb{R}^3$, pinned by P1 verbatim. The frame map is Eq. (2) applied componentwise. Because Galilean boosts commute and do not rotate spin, composition, inverses, the rest of partners (a $w < 1$ partner at the fraction $1 - w$ of its relative velocity) and the laboratory pinning hold exactly as in one dimension — verified on five particles (two pairs, of weights 1 and 0.6, and an unrelated fifth) over all frame triples and all pinned heights.

Covariant form. Let the potential be group-valued: $\Lambda_k(z) \in \text{SO}^+(1, 3)$ maps laboratory coordinates to frame k 's at height z , pinned by the same rule — at a partner height z_j , $\Lambda_k(z_j) = B(\mathbf{u}_{kj}) B(\mathbf{v}_j)^{-1}$, where $B(\mathbf{v})$ is the

pure boost to velocity $\mathbf{v}$ and $\mathbf{u}_{kj}$ is $(1 - w_{kj})$ times the relativistic relative velocity of j in k 's rest frame (an object at rest after $B(\mathbf{v}_j)^{-1}$ moves at $-\mathbf{u}_{kj}$ after $B(\mathbf{u}_{kj})$; the magnitude is the statement, the sign a convention). The frame map is the right quotient $T_{kl}(z) = \Lambda_l(z) \Lambda_k(z)^{-1}$, so $T_{lm} \circ T_{kl} = T_{km}$ and $T_{lk} = T_{kl}^{-1}$ hold identically, by group multiplication, for **any** assignment of group elements — verified over all frame pairs and triples and all heights of a four-particle configuration with random non-collinear velocities up to $0.79c$ (sampler capped at $0.83c$) (two pairs of weights 1 and 0.4), at the 10^{-10} level. A $w = 1$ partner sits at $\mathbf{x}' = 0$ with frame time equal to its proper time since creation; a $w < 1$ partner moves at $(1 - w)$ times the relativistic relative velocity; for collinear velocities every rotation part vanishes, recovering Eq. (8). For non-collinear velocities the maps between two unrelated particles' frames are not pure boosts: their Wigner rotations reach $\approx 27^\circ$ in the scan just described (over all maps between different particles' frames at all pinned heights; 18° between the frames of particles in different components); the value is scan-specific, two perpendicular $0.85c$ boosts alone giving 34° . On the sharp tracks of Sec. 3.5 each such rotation acts on a particle's spin as a fixed local unitary, which changes no cut negativity, no edge weight and no laboratory statistic; what it does change is the orientation convention at each pinned height — the $SO(3)$ gauge priced in Sec. 3.7.4.

The interval clock. Eq. (9) is dimension-free once the spatial displacement is read with the Euclidean norm: the signed interval from the defining event is a Lorentz scalar in any dimension and its sign is cone membership. Over 300 random two-pair configurations (random defining event, measurement times and velocities up to $0.5c$) and random non-collinear boosts up to $0.85c$, both clocks, the sign of each and their order are invariant at the 10^{-9} level; and the order-disagreement fraction $(R - 1)/(2R)$ of Fig. 3 holds unchanged for perpendicular velocities, since only the γ factors enter.

Photons. On a null worldline from the defining event the interval clock reads zero, so every photon pair born at that event is on the tie set of Eq. (9). The affine parameter a photon's own objects supply does not share the defect: for two photons flying apart from one event, measured at laboratory times 1.0 and 1.3 (arbitrary units), $\lambda = t/\omega$ is invariant to 10^{-15} under boosts $\beta \in (-0.9, 0.9)$ along the flight axis and its order flips under 0 of 181 of them — each photon's coordinate time and frequency rescale by the same Doppler factor $\sqrt{(1 \mp \beta)/(1 \pm \beta)}$, and only a parameter normalized by laboratory time (which flips under 77 of 181) is frame-dependent. This is the “ordered but not located” statement of Sec. 3.7.4: what photons lack is a rest frame, not a clock.

Every claim in this appendix is verified in the released code (Table 5).

Code and data availability

Every “verified numerically” in this paper is backed by released code: a Python/NumPy package implementing the frame maps, the locality-graph and edge-weight rules, the covariant variant and its ordering rule, the contact-rule enumeration and the propositions' numerical checks, with a unit-test suite in which each numbered claim has a test (Table 5). The package, its tests and the scripts that generate every figure of this paper and the numerical tables moved to the repository are publicly available under the MIT licence at https://github.com/verlyn-texas/entanglement_as_locality_verification_code and archived at [doi:10.5281/zenodo.22820042](https://doi.org/10.5281/zenodo.22820042) (release v1.2.2, the version described here). The paper uses no data beyond the published values it quotes.

Declarations

This work received no external funding. The author declares no competing interests. The manuscript is not under consideration elsewhere.

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